Public or Private? A Unified Demand–Supply Coupled Framework for EV Infrastructure Investment

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ABSTRACT

Utilizing taxpayer dollars for electric vehicle (EV) charger deployments has sparked ongoing debate over the efficiency and fairness of public spending. Public agencies aim to maximize utilization and accessibility, whereas private investors prioritize profitability. This study develops a unified modeling framework that represents public-led and private-led charging station deployment as heterogeneous decision processes and solves it using a bilevel demand-supply coupling approach. The upper level compares public-led, private-led, and coexistence scenarios under a fixed government budget, while the lower level predicts station-level utilization. The two levels are iteratively coupled: supply-side siting decisions shape demand forecasts, while demand feedback updates supply through a gradient-informed spatial expansion algorithm. Results show that private-led deployment can leverage matching capital to improve utilization and profit, but may increase user charging costs and bypass some lower-income minority areas. Sensitivity analyses indicate that private-led profitability is robust to EV adoption growth but depends strongly on charging-to-purchase conversion.

Keywords: Electric vehicle; Charging station location problems; Public-private investment; Bi-level optimization; Demand-supply coupling

1.Introduction

Recent federal initiatives have increasingly emphasized the need for efficiency gains and spending reductions, especially in large-scale infrastructure investments. In the context of electric vehicle (EV) studies, the use of taxpayer dollars to deploy charging infrastructure—illustrated by controversies surrounding programs such as the National Electric Vehicle Infrastructure (NEVI) initiative—has sparked intensified debate over the effectiveness and equity of public spending (Zraick & Hubler, 2025). However, despite rapid technological progress, the lack of adequate chargers continues to impede broader adoption. Empirical evidence shows that a 1% increase in charging accessibility can drive a 5.75% rise in EV uptake (Qian et al., 2024), underscoring the importance of infrastructure availability. Given the substantial capital costs and the immobility of fixed infrastructure, poor siting can result in underutilized and expensive assets that are difficult to relocate (Jochem et al., 2016). Consequently, enhancing government efficiency by promoting the more effective use of taxpayer dollars and quantitatively evaluating impacts on community outcomes emerges as a timely and essential research imperative.

The market mechanism has consistently demonstrated its effectiveness in efficiently allocating resources, leveraging competition to ensure that capital is directed toward its most productive uses. This principle, deeply rooted in classical economic thought, was originally articulated that "It is not from the benevolence of the butcher, the brewer, or the baker, that we expect our dinner, but from their regard to their own interest" (Smith, 1937). Applied to EV infrastructure, this logic gives rise to two distinct investment strategies:



Figure 1. Two distinct investment strategies: (a) Public free charger located in front of a municipal building; (b) Private charger situated in front of a shopping mall

- **Public agency-led investment (or public-led, for short)**: Public agencies, such as townships or municipalities, assume full financial responsibility, offering charging services at minimal or no cost. An example of a public charger situated in front of a municipal building, offering free charging to local residents and visitors is shown in Figure 1-a.

- **Private-led investment (also referred to as business-led)**: Private businesses can and often do serve as charging infrastructure owner operators (IOOs). Drawing on their decentralized structure, much like their involvement in other utility sectors such as power grid management or waste collection, private IOOs install EV chargers to attract drivers and encourage longer visits (see an private charger example situated in front of a shopping mall in Figure 1-b). Research shows that

business owners—whether they run restaurants, convenience stores, grocery stores, or shopping centers—can earn between $1 and $50 in profit per customer visit with a purchase, depending on the type of business and various other factors (NRAAssociation, 2025; IBISWorld, 2024, 2025a, 2025b; NACSStores, 2023).

These two investment models are not entirely separate; rather, they often involve some degree of coordination between public agencies and private businesses. Public agencies frequently retain influence over private IOOs' decision-making processes, especially when private entities seek public subsidies (e.g., through "Driving PA Forward Grant and Rebate" program (DEP, 2025)) to offset their initial investment costs. In such scenarios, agencies typically employ a proposal selection process, reviewing and evaluating requests from various private IOOs. This selection mechanism allows public authorities to strategically influence charger deployment, guiding private investments to align with broader public interests such as balanced service coverage and optimal network layout.

Despite the prevalence of such coordination in practice, existing CSLP formulations rarely represent the decision problem in which a public agency allocates a fixed funding budget across alternative deployment under distinct objectives. Specifically, if a federal agency has a fixed budget F, how should it distribute funds between direct public investment and subsidies for private deployment? The budget may be allocated entirely to public-led projects ($F^p = F$ where p stands for public), entirely to private-led projects ($F^b = F$ where b stands for private), or split between the two ($F^p + F^b = F$). This decision is not a conventional single-planner siting problem because public agencies, private investors, users, and existing charging stations are associated with different objectives, constraints, and performance outcomes. Therefore, a unified framework is needed to represent the real-world public-private deployment process and evaluate its implications for utilization, profitability, and user cost. Developing such a framework involves three technical challenges:

- First, the model needs to represent the divergent objectives and constraints of public-led and private-led deployment within a common decision structure. Public agencies emphasize public-fund efficiency, utilization, and accessibility, whereas private investors emphasize profitability, investment feasibility, and business-related revenue.
- Second, the model needs to capture how deployment strategies reshape station-level utilization across both new and existing chargers. Because new chargers alter local supply conditions and may redistribute demand among nearby stations, utilization cannot be treated as fixed or independent of siting decisions.
- Third, the model needs to integrate stakeholder objectives and utilization responses through a demand-supply coupled decision process. Predicted utilization affects the marginal value of candidate sites, while deployment decisions update the supply configuration that shapes future utilization.

To address these challenges, this paper makes two main contributions. First, it develops a unified CSLP formulation that represents public-led and private-led deployment as heterogeneous decision processes with different objectives, constraints, and performance outcomes. The formulation evaluates three scenarios,

public-led, private-led, and public-private coexistence, and explicitly captures cross-actor competition at shared candidate sites. Second, it extends the demand–supply coupling structure, established for a single planning entity in prior CSLP work, to this public-private setting, where station-level utilization is predicted in response to competing siting decisions and then used to guide subsequent supply-side updates. The resulting coupled optimization problem is solved by a gradient-informed spatial expansion (GISE) algorithm.

The proposed framework consists of three connected components. The upper-level model determines charger siting under the three investment scenarios. In the public-led scenario, the objective is to maximize cumulative charging utilization under public funding and service-density constraints. In the private-led scenario, the objective is to maximize profit, including charging revenue, parking or overstay fees, and ancillary business revenue. In the coexistence scenario, public and private actors may compete for overlapping candidate sites, and site assignment is resolved based on their marginal contributions to utilization and profit.

The lower-level model predicts station-level utilization for both existing and candidate chargers using a gradient boosting-based ensemble learning model trained on historical charging event data. The model incorporates contextual predictors such as land use, station density, and traffic volume to estimate localized charging demand.

Finally, the upper- and lower-level models are linked iteratively. Upper-level siting decisions update the spatial supply configuration, while lower-level utilization predictions provide demand feedback for evaluating the marginal value of each site. The GISE algorithm uses this feedback to update the deployment plan until it stabilizes.

The remainder of this paper is organized as follows. Section 2 reviews the existing literature on charger location modeling and public–private investment strategies. Section 3 details the development of the bi-level optimization framework, presents the corresponding mathematical formulations, and introduces the proposed solution algorithm. Section 4 applies the model to a real-world case study, including performance evaluation and sensitivity analyses. Finally, Section 5 concludes with key findings, policy implications, and suggestions for future research.

2. Literature Review

The charging station location problem (CSLP) has recently emerged as a popular research topic within transportation infrastructure planning, as global EV adoption accelerates and the need for effective charging networks becomes increasingly critical. The conceptual basis for CSLP originates from general facility location theory, which traditionally focused on optimally positioning facilities to serve stationary demand at discrete points. Klose and Drexl (2005) have conducted a comprehensive review regarding such approaches. Recognizing the limitations of node-based formulations for transportation systems, researchers developed flow-based models that consider vehicle movements along network paths (Hodgson, 1990; Kuby & Lim, 2005). These foundational studies, while originally focused on general refuel infrastructure planning, have informed the methodological development of CSLP research. Following the early

conceptual foundation of general refueling station location models, research on CSLP has grown steadily as a distinct area of study. Initial CSLP studies extended classical facility location planning methods by incorporating EV-specific constraints such as vehicle range limits and grid capacity considerations (Liu et al., 2012; Wang & Lin, 2009). As research has deepened, studies have shown four trends: a progression from single objective to multi-criteria models (Baouche et al., 2014; Loni & Asadi, 2023); a transition from static to dynamic planning (Kadri et al., 2020; Li et al., 2016); a transition from single-level model to multi-level model (Huang & Kockelman, 2020; Song & Hu, 2023; Zeng et al., 2020); and an evolution from spatial layout optimization to more comprehensive approaches integrating behavioral characteristics, grid coordination, and system efficiency (Cui et al., 2019; Pal et al., 2023; Wang et al., 2019).

In the real world, public and private EV charger deployments differ significantly in terms of decision-making motivations, spatial distribution, and operational characteristics. Public chargers, typically managed by municipalities, townships, or other public agencies, are more evenly distributed across urban and suburban areas to promote accessibility and equity (He et al., 2022). On the contrary, private charging networks tend to cluster in commercial centers, shopping malls, and employment hubs, reflecting the operators' focus on customer access and business opportunities. Betz et al. (2017) indicated that corporate charging facility deployment decisions are primarily influenced by corporate goals, available parking, and customer retention benefits, rather than network equity. Flinders and Brusseler (2024) further observed that a market-led approach has led private operators to prioritize areas with high demand and strong returns, contributing to significant geographic disparities in provision.

Reflecting this divergence in objectives and deployment patterns, some studies have developed planning models for private chargers, addressing issues such as profitability and energy loss trade-offs (Firouzjah, 2022), incentive policy impacts (Fang et al., 2020), and private investment responses to demand guarantees and cost-sharing schemes (Kizhakkan et al., 2019). In addition, research has examined interactions among multiple stakeholders, including those between public and private actors. For example, Wang et al. (2025) analyze the interactions among the government, charging platforms, and private charger owners, exploring how these stakeholders extract value from private chargers. Yu et al. (2025) and Song et al. (2021) focus respectively on the impacts of government subsidies on EV automakers versus fuel vehicle automakers, and on electricity price regulation during the operating stage. However, their primary focus differs from the CSLP problem addressed in this study. By contrast, most existing CSLP models either assume a single planning entity with unified objectives, or treat charger deployment under a homogeneous ownership structure. Such assumptions simplify the fragmented nature of real-world infrastructure deployment, where governments and private companies operate independently under diverse objectives and regulatory frameworks. Consequently, the coexistence and interaction of multiple actor and charger types within a unified planning model remain insufficiently represented in current CSLP models.

Regarding the siting methodology, CSLP approaches are generally categorized into flow-based and node-based models (Kchaou-Boujelben, 2021). Flow-based models represent charging demand through origin-destination (O-D) trip flows and locate chargers along routes to allow drivers to complete round trips without exhausting vehicle range (Li et al., 2022; Wang & Lin, 2009; Yıldız et al., 2016). A key limitation of this approach is that it relies on the availability of accurate EV O-D demand matrices, which depend on

detailed data regarding EV ownership, trip frequency, and trip destinations—data that are often unavailable or difficult to obtain for planners. In contrast, node-based models assume that charging demand occurs at specific nodes representing home, work, or POIs, and estimate demand based on the size of the station's service area (Cavadas et al., 2015; Huang et al., 2016; Klose & Drexl, 2005; Zhou et al., 2022). This introduces another limitation, as demand is indirectly inferred by generalized spatial coverage rather than actual usage. Although these limitations are most apparent in node-based models, flow-based approaches also rely on surrogate measures such as flow volumes or trip counts, rather than direct station-level usage data. As a result, most existing charger planning models rely on indirect proxy metrics to estimate demand, such as distance to major roads, service area size, or generalized traffic volumes. While these proxies offer some guidance, they do not capture station-level performance metrics such as occupancy rates or the number of daily charging events.

Another key yet underexplored challenge in CSLP research is the lack of integrated demand-supply dynamic coupling in station planning models. Most existing studies treat charging demand as an exogenous and static input, independent of infrastructure decisions. Some works have attempted partial behavioral integration, such as Huang and Kockelman (2020), who incorporated congestion and price feedbacks into a station location-routing bilevel model, and Zeng et al. (2020), who modeled EV user responses to pricing decisions within a bilevel optimization framework. Additionally, recent approaches have also leveraged large-scale datasets, including vehicle trajectories, travel surveys, and charging transaction records, combined with machine learning algorithms, ensemble learning, and simulation-based forecasting, to predict charging needs under varying adoption and policy scenarios (Arias et al., 2017; Wang et al., 2023; Yi et al., 2022; Zhang et al., 2023). While these learning-based methods offer valuable inputs for infrastructure planning, they remain focused solely on demand forecasting and are not integrated within CSLP frameworks for simultaneous station location and demand estimation. Song and Hu (2023) proposed an iterative two-stage framework linking station-level demand prediction with siting, but under a single-planner assumption. Extending dynamic coupling to a multi-actor planning framework with distinct objectives remains open.

To summarize, we identify one principal gap and one related methodological gap. The principal gap is that existing CSLP models assume a single decision-maker with unified objectives or a homogeneous ownership structure, failing to capture the coexistence and competition of public and private deployment under divergent objectives. The related methodological gap is that demand–supply coupling in CSLP remains confined to single-planner settings and has not been fully extended to multi-actor frameworks. Addressing these gaps is essential for developing more realistic and adaptable tools to guide EV charging infrastructure deployment in complex, dynamic environments.

3. Methodology

3.1. Overview of the bi-level optimization framework

In this section we present a bi-level optimization framework for EV charger deployment, as shown in Figure 2. The central question addressed is: given a fixed total funding amount F, (1) how should a government agency allocate resources between direct public investment and private-sector subsidies, and (2) how do

the resulting outcomes differ for residents and businesses? The funding allocation can take one of three forms—fully public-led deployment ($F^p = F$), fully private-led deployment ($F^b = F$), or a mixed deployment where $F^p + F^b = F$, with p representing public and b representing private(business).

To operationalize this framework, we begin with the upper-level (Section 3.2), where we model how public-led and private-led investment strategies solve CSLP based on their respective objectives and decision-making processes. In the upper-level model, the potential charging utilization at each candidate site, represented by occupancy, is initialized based on site-specific characteristics and temporarily treated as an exogenous parameter for the purpose of infrastructure layout optimization. Its dependence on the overall infrastructure configuration and traveler behavior is explicitly modeled and endogenously updated in the lower-level model. We then formulate the objective functions, constraints, and performance metrics specific to each investment scenario. In the lower-level (Section 3.3), we present a boosting-based ensemble learning method to predict station-level charging utilization. Here, the infrastructure layout decisions from the upper-level are treated as inputs, and we use a real-world charging dataset collected from Kansas City, Missouri, to train the demand prediction model. The lower-level model then explicitly captures the dependence of charging utilization on the deployed infrastructure configuration and traveler behavior, and generates updated occupancy estimates for all candidate sites under the proposed deployment. Section 3.4 describes the solution algorithms used to iteratively solve the coupled bi-level framework until convergence is achieved, that is, when the outputs of both levels, occupancy estimates and siting decisions, stabilize across successive iterations.

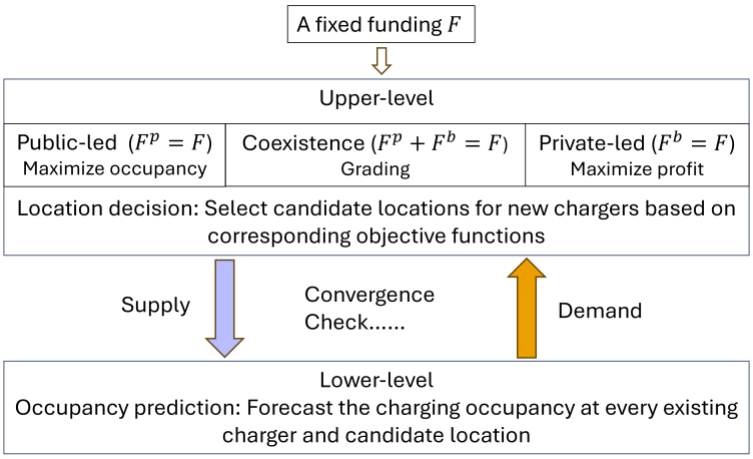


Figure 2. Schematic of the bi-level optimization framework

Upon convergence, we compare outcomes across different strategies using the two key indicators: the *total occupancy* and the *total profit* of all chargers deployed in the network. Total occupancy is defined as the sum of the usage durations of all deployed chargers over a specified period (e.g., a month). It serves as a practical indicator of infrastructure effectiveness, reflecting both service quality in public deployments and utilization levels relevant to private profitability. Total profit, though not a direct objective for public chargers, is still calculated to account for potential long-term financial liabilities (if profit is a negative number) and to enable cross-scenario comparisons. Finally, it is important to note that both newly planned investments and existing charging infrastructure are considered when calculating these two indicators. As

extensively discussed in the literature, traveler behavior is influenced by the availability and spatial distribution of charging options. Therefore, the deployment of new chargers inevitably alters the usage patterns of existing stations. Modeling this interaction ensures a more realistic representation of system-wide impacts and better supports infrastructure planning decisions.

For easy reference purposes, the notations used in this study are summarized in Table 1.

Table 1 Definition of notations in upper-level model

Notation	Definition
p, b	Superscript. Public-led and private-led (aka business-led) chargers, respectively
e, c	Subscript. Existing and newly constructed chargers, respectively
i, j	Subscript. Index of charger locations
t, T	Subscript. Index of the year (the t-th year), within an investment period of T years
m, M	Subscript. Index of the month (the m-th month), within a data temporal scale (low-level model training) or investment period (prediction) of M months
x_i^p, x_i^b	Binary variables, whether location i is chosen to build a public or private charger, respectively
y_j^i	Binary variable, whether two locations, i and j, are in the same area
N	Number of chargers. When used with superscript and subscript, N may take the forms of N_e^p, N_e^b, N_c^P, N_c^b
I	Location set of chargers. It may take the forms of I_e^p, I_e^b, I_c^p, I_c^b
ρ	Occupancy of chargers (hours), it may take the forms of $\rho_{i,t}^p, \rho_{i,t}^b$, $\rho_{i,m}$, ρ_e^p, ρ_e^b, ρ_c^p, ρ_c^b
Π	Profit of chargers. It may take the forms of $\Pi_{i,t}^p, \Pi_{i,t}^b$, Π_e^p, Π_e^b, Π_c^p, Π_c^b
R	Revenue of chargers, it may take the forms of $R_{i,t}^p, R_{i,t}^b$
P	Overstay parking duration (hours), it may take the forms of $P_{i,t}^p, P_{i,t}^b$
E	Energy consumption (kWh), it may take the forms of $E_{i,t}^p, E_{i,t}^b$
Q	Number of charging sessions (times), may include $Q_{i,t}^p, Q_{i,t}^b$
s, ε, u	Service fee per kWh, electricity fee per kWh, and overstay parking fee per hour
$\tau, \vartheta, \bar{P}$	Charging efficiency, overstay parking-to-occupancy ratio and average overstay parking duration per session
G, C, O	One-time government subsidy, construction cost, and annual operating cost for chargers
F, F^P, F^b	Total government funds, and their allocations to public and private chargers, respectively.
r	Discount rate per year
ω	The ratio of annual additional operating costs, excluding electricity cost, relative to the construction cost.
$\pi_{i,t}$	Median profit per in-store transaction for the private business co-located with a private charger i
$\alpha_{i,t}$	Probability of using a private charger i and making a purchase in co-located business
$\beta_{i,t}$	Discount coefficient when using a private charger i and making a purchase (if available)
B_i	Private investor's budget for a new private charger located at i
$D_i, N_i, A_i,$	Density and number of chargers of the same ZIP code area to which the charger located at i belongs, and the size of that area
$a_{i,m}$	Feature vector used in lower-level model
ζ	Compound annual growth rate (CAGR) for EV adoption

3.2. Upper-level CSLP optimization model for charging infrastructure planning

This subsection focuses on the supply side of the planning problem, where we develop distinct optimization models to address the CSLP under three investment scenarios: public-led, private-led, and public–private coexistence. These scenarios are detailed in the following three subsections, each formulated to reflect the specific objectives and decision-making logic of the corresponding investment strategy. In the upper-level model, driver behavior is assumed to be fixed. For a given iteration, if a candidate site is selected for charger installation, its occupancy level ρ is treated as a provisional input reflecting the site's standalone utilization potential, independent of other siting decisions. We then formulate the objective functions, constraints, and performance metrics specific to each investment scenario. Once these siting decisions are made, they are passed to the lower-level model, where traveler behavior is allowed to adapt and occupancy levels across all sites are jointly updated in response to the resulting infrastructure layout.

3.2.1. CSLP model formulation for the public-led investment strategy

In this section, we formulate the CSLP model as a binary optimization model for public-led investment strategy. We assume there are N_e^p existing public chargers, and the government plans to allocate the entire fund, $F^p = F$, to construct additional public chargers at a subset of the N_c^p available candidate locations. We define two location sets: I_e^p, representing existing public charger locations, with $|I_e^p| = N_e^p$; and I_c^p, representing candidate locations for new public chargers, with $|I_c^p| = N_c^p$. A binary decision variable x_i^p is used to indicate whether location i either hosts an existing charger or is selected for the construction of a new public charger, as shown in Eqs. (1)-(2).

$$x_i^p = 1, \text{ for } i \in I_e^p \tag{1}$$

$$x_i^p = \begin{cases} 1, & \text{if } i \text{ is selected to build a public charger} \\ 0, & \text{otherwise} \end{cases}, \text{ for } i \in I_c^p \tag{2}$$

As discussed in Section 1, the objective of a public-led strategy is to maximize overall charging utilization. These chargers are typically deployed in high-demand areas, such as along freeways or near major community centers. To reflect this planning principle, we use occupancy $\rho_{i,t}^p$—defined as the total duration a public charger i is in use within a specified period t—as a practical indicator of infrastructure effectiveness. Occupancy captures both the service quality of public deployments and the utilization levels relevant to private-sector profitability. Considering both newly constructed and existing charger sets, the total occupancies ρ_c^p and ρ_e^p for new and existing public chargers, respectively, are given by Eqs. (3)-(4).

$$\rho_c^p = \sum_{i \in I_c^p} x_i^p \sum_{t \in T} \rho_{i,t}^p \tag{3}$$

$$\rho_e^p = \sum_{i \in I_e^p} \sum_{t \in T} \rho_{i,t}^p \tag{4}$$

To evaluate the performance of the entire network rather than focusing solely on new chargers, we use $\rho_{e,cur}^p$ to denote the total occupancy of existing public chargers prior to optimization. The change in occupancy of existing chargers before and after optimization, denoted as $\Delta\rho_e^p$, is then calculated as shown in Eq. (5).

$$\Delta\rho_e^p = \rho_e^p - \rho_{e,cur}^p \quad (5)$$

By defining the objective function as the sum of the total occupancy of new chargers and the change in occupancy of existing chargers (Eq. (6)), we aim to maximize overall system occupancy Φ^p.

$$max\ \Phi^p = \rho_c^p + \Delta\rho_e^p \quad (6)$$

We also model the metric Π to represent the profit of chargers. Although profit is not a direct objective for public chargers, it is still calculated to account for potential long-term financial liabilities (if profit is a negative number) and gains, and to facilitate cross-scenario comparisons. For a public charger at location i, the profit $\Pi_{i,t}^p$ is defined as the revenue minus the costs incurred in year t.

For the public-led model, the revenue $R_{i,t}^p$ includes income from charging service fees, electricity fees, and overstay parking fees, as expressed in Eq. (7). In practice, not all of these fees may be charged simultaneously. However, to ensure the model's completeness and practical relevance, we include all potential fees that users might be required to pay. This comprehensive approach allows the model to accommodate diverse pricing structures and regulatory environments, enhancing its adaptability to real-world applications.

$$R_{i,t}^p = E_{i,t}^p(s_{i,t}^p + \varepsilon_{i,t}^p) + P_{i,t}^p \cdot u_{i,t}^p \quad (7)$$

where $E_{i,t}^p$ is energy consumption (in kWh). $s_{i,t}^p$ and $\varepsilon_{i,t}^p$ are the service fee and electricity fee per kWh. $P_{i,t}^p$ and $u_{i,t}^p$ are the overstay parking duration (in hours) and overstay parking fee per hour.

Both $E_{i,t}^p$ and $P_{i,t}^p$ characterize charger-specific charging behavior realized during year t, and are empirically related to the charger's occupancy level $\rho_{i,t}^p$. Based on the observed charging records, each charger exhibits relatively stable operational patterns over a given period, which can be approximately quantified through proportional relationships with occupancy. Specifically, charging efficiency, measured in kWh per hour, is recorded in the dataset, allowing us to extract patterns over time. Let τ_i^p denote the charging efficiency. Then energy consumption is calculated as Eq. (8).

$$E_{i,t}^p = \tau_i^p \cdot \rho_{i,t}^p \quad (8)$$

Similarly, the ratio of overstay parking duration to occupancy is observed in the data and reflects typical post-charging parking behavior at a given charger. Let ϑ_i^p denote this ratio. Then the overstay parking duration is calculated as Eq. (9).

$$P_{i,t}^p = \vartheta_i^p \cdot \rho_{i,t}^p \quad (9)$$

Although ϑ_i^p is typically close to 1, drivers do not always move their vehicles immediately after charging completes, leading to additional idle time at the station. This phenomenon—commonly observed in real-world datasets—results in $\vartheta_i^p \geq 1$. We emphasize that ϑ_i^p is not imposed as a universal behavioral constant, but rather serves as a charger-specific empirical parameter capturing average overstay behavior conditional on realized occupancy during the period. Variations in occupancy driven by infrastructure layout and station density are explicitly modeled in the lower-level formulation, while the proportional

relationship provides a parsimonious representation of station-level behavioral patterns within each iteration.

We next consider the costs associated with building and operating a public charger. For public-led chargers, the agency assumes full financial responsibility for construction and must cover the entire cost. The annual operating cost $O_{i,t}^{p}$ primarily includes electricity cost and additional operating costs, which can be expressed as Eq. (10).

$$O_{i,t}^{p} = E_{i,t}^{p} \cdot \varepsilon_{i,t} + C_{i}^{p} \cdot \omega_{i,t}^{p} \tag{10}$$

where $\varepsilon_{i,t}$ is the electricity cost per kWh that the charger owners must pay to the utility company. C_{i}^{p} is the one-time construction cost. $\omega_{i,t}^{p}$ is the ratio of annual additional operating costs relative to the construction cost C_{i}^{p}.

Within the defined investment period, the profits from new and existing chargers—denoted as Π_{c}^{p} and Π_{e}^{p}, respectively—as well as the profit change for existing chargers, $\Delta\Pi_{e}^{p}$, are defined in Eqs. (11)-(13).

$$\Pi_{c}^{p} = \sum_{i \in I_{c}^{p}} x_{i}^{p} \left(\sum_{t=1}^{T} \frac{R_{i,t}^{p} - O_{i,t}^{p}}{(1+r)^{t}} - C_{i}^{p} \right) \tag{11}$$

$$\Pi_{e}^{p} = \sum_{i \in I_{e}^{p}} \sum_{t=1}^{T} \frac{R_{i,t}^{p} - O_{i,t}^{p}}{(1+r)^{t}} \tag{12}$$

$$\Delta\Pi_{e}^{p} = \Pi_{e}^{p} - \Pi_{e,cur}^{p} \tag{13}$$

For newly constructed chargers, the construction cost C_{i}^{p} must be included in the analysis, whereas for existing chargers, this cost is excluded. In Eqs. (11)-(12), r denotes the discount rate per year. To ensure methodological rigor, we fully incorporate the concept of discounting from finance, accounting for the time value of money in calculating profits over multiple years. With these considerations, the secondary performance metric for public chargers is defined in Eq. (14).

$$\Pi^{p} = \Pi_{c}^{p} + \Delta\Pi_{e}^{p} \tag{14}$$

Next, Eq. (15) represents the funding constraint, indicating that the government has a fixed funding F^{p} to support both the construction and operation of public chargers. The operation of public chargers may result in either a profit or a loss, which is captured through the inclusion of $(O_{i,t}^{p} - R_{i,t}^{p})$ in the equation. In some cases, government funding for construction and operation is allocated separately. However, since both are considered part of infrastructure investment, this constraint accounts for both types of expenditures simultaneously.

$$\sum_{i \in I_{c}^{p}} x_{i}^{p} \left(\sum_{t=1}^{T} \frac{O_{i,t}^{p} - R_{i,t}^{p}}{(1+r)^{t}} + C_{i}^{p} \right) \leq F^{p} \tag{15}$$

3.2.2. CSLP model formulation for the private-led investment strategy

In this section, we formulate the CSLP model also as a binary optimization model for private-led investment strategy. We assume there are N_e^b existing chargers, and the government plans to allocate the entire fund, $F^b = F$, as subsidies for business owners to construct additional private chargers at a subset of the N_c^b available candidate locations. We define two location sets: I_e^b, representing existing charger locations, with $|I_e^b| = N_e^b$; and I_c^b, representing candidate locations for new chargers, with $|I_c^b| = N_c^b$. A binary decision variable x_i^b is used to indicate whether location i either hosts an existing charger or is selected for the construction of a new charger, as shown in Eqs. (16)-(17).

$$x_i^b = 1 \text{ for } i \in I_e^b \tag{16}$$

$$x_i^b = \begin{cases} 1, & \text{if } i \text{ is selected to build private chargers} \\ 0, & \text{otherwise} \end{cases}, \text{ for } i \in I_c^b \tag{17}$$

The objective of deploying private chargers is to maximize profit—that is, to maximize the difference between total revenue and associated costs.

For a private charger located at location i, the revenue in year t, denoted $R_{i,t}^b$, includes components similar to those of a public charger, such as charging service fees, electricity charges, and overstay parking fees. In addition, private chargers generate business income (Eq. (18)), which refers to the revenue from customer purchases made at the host business while their vehicles are charging. As with public charging operations, not all revenue components are necessarily collected simultaneously. The specific combination depends on the pricing strategy adopted by the private investor.

$$R_{i,t}^b = E_{i,t}^b(s_{i,t}^b + \varepsilon_{i,t}^b) + P_{i,t}^b \cdot u_{i,t}^b + Q_{i,t}^b \cdot \pi_{i,t} \cdot \alpha_{i,t} \cdot \beta_{i,t} \tag{18}$$

where the first two terms follow the public revenue in Eq. (7). The third term is unique to private-led chargers and represents additional business income. $Q_{i,t}^b$ is the number of charging sessions, which is computed using equation (20). $\pi_{i,t}$ is the median profit per in-store transaction, which varies by business type (NRAAssociation, 2025; IBISWorld, 2024, 2025a, 2025b; NACSStores, 2023). $\alpha_{i,t}$ represents the probability that a customer who charges their vehicle also makes a purchase at the host business. $\beta_{i,t}$ is the discount coefficient, which reflects promotional incentives linked to charger use and in-store purchases.

The values of $E_{i,t}^b$ and $P_{i,t}^b$ are derived from occupancy $\rho_{i,t}^b$, using formulas similar to those used for public chargers, as shown in Eq.(19), where τ_i^b is the average charging efficiency and ϑ_i^b is the ratio of overstay parking duration to occupancy.

$$E_{i,t}^b = \tau_i^b \cdot \rho_{i,t}^b; \; P_{i,t}^b = \vartheta_i^b \cdot \rho_{i,t}^b \tag{19}$$

Since individual session's energy consumption is recorded, the average energy consumption per session, $\bar{E}_i^b$, can be obtained. This enables the calculation of $Q_{i,t}^b$, as follows.

$$Q_{i,t}^b = E_{i,t}^b / \bar{E}_i^b \tag{20}$$

On the cost side, we use C_i^b and $O_{i,t}^b$ to denote the one-time construction cost and the annual operating cost for year t (Eq. (21)), respectively—similar to the public charger model. Generally, $\varepsilon_{i,t}^b$ is set to be

greater than $\varepsilon_{i,t}$ in order to ensure profitability.

$$O_{i,t}^{b} = E_{i,t}^{b} \cdot \varepsilon_{i,t} + C_{i}^{b} \cdot \omega_{i,t}^{b} \tag{21}$$

Besides, for private chargers, we introduce a new variable, G_i, representing a one-time government subsidy that private businesses may receive to offset construction costs. Such subsidies may be available through programs like the Charging and Fueling Infrastructure Discretionary Grant Program provided by the U.S. Department of Transportation. Let B_i represent the private investor's own capital allocated for constructing a charger at location i. Given these definitions, the following equations govern the private investment model:

$$B_i + G_i \geq C_i^b \tag{22}$$

$$\Pi_i^b = \sum\nolimits_{t=1}^{T} \frac{R_{i,t}^b - O_{i,t}^b}{(1+r)^t} - B_i \geq 0 \tag{23}$$

$$\Pi_c^b = \sum_{i \in I_c^b} x_i^b \cdot \Pi_i^b \tag{24}$$

$$\Pi_e^b = \sum_{i \in I_e^b} \sum\nolimits_{t=1}^{T} \frac{R_{i,t}^b - O_{i,t}^b}{(1+r)^t} \tag{25}$$

$$\Delta\Pi_e^b = \Pi_e^b - \Pi_{e,cur}^b \tag{26}$$

As previously discussed, investing in private chargers is treated analogously as a business venture. Before a new charger is deployed at location $i \in I_c^b$, it is necessary to evaluate both the sufficiency of available capital and the projected profitability. These conditions are represented by the budget feasibility constraint (Eq. (22)) and the business feasibility constraint (Eq. (23)). Specifically, Eq. (22) states that the combined value of the private investor's budget B_i and the government subsidy G_i must meet or exceed the required construction cost C_i^b. This highlights how public subsidies can significantly lower the capital threshold for private investors. Eq. (23) ensures that the net present value of future profits over the planning horizon, discounted at rate r, must exceed the investor's contribution B_i. This again demonstrates the role of G_i in relaxing financial barriers and increasing investor confidence.

Eq. (24) defines the cumulative profit from newly constructed private chargers, while Eq. (25) calculates the profit from existing private chargers. Eq. (26) captures the change in profit for existing private chargers resulting from the optimized deployment.

Finally, Eq. (27) defines the private-led model's objective function: to maximize the sum of profits from both new and existing private chargers.

$$max\ \Pi^b = \Pi_c^b + \Delta\Pi_e^b \tag{27}$$

The secondary evaluation metrics for private chargers are presented in Eq. (31), using a structure analogous to that of public chargers (see Eqs. (28)-(30)).

$$\rho_c^b = \sum_{i \in I_c^b} x_i^b \sum_{t \in T} \rho_{i,t}^b \tag{28}$$

$$\rho_e^b = \sum_{i \in I_e^b} \sum_{t \in T} \rho_{i,t}^b \tag{29}$$

$$\Delta\rho_e^b = \rho_e^b - \rho_{e,cur}^b \tag{30}$$

$$\Phi^b = \rho_c^b + \Delta\rho_e^b \tag{31}$$

Here, ρ_c^b represents the cumulative occupancy of newly constructed private chargers, while ρ_e^b denotes the total occupancy of existing private chargers. The change in occupancy at existing sites due to optimization is captured by $\Delta\rho_e^b$. Eq. (31) defines the overall occupancy-based performance metric for the private deployment scenario.

Similar to the public charger model, the private charger model incorporates a funding constraint—referred to as the subsidy constraint (Eq. (32))—which limits the total amount of government subsidies allocated to private charger deployment to a predetermined budget F^b.

$$\sum_{i \in I_c^b} x_i^b \cdot G_i \leq F^b \tag{32}$$

3.2.3. CSLP model formulation for the public-private coexistence investment strategy

In the public–private coexistence scenario, the challenge lies in determining how different allocations of a total funding amount F between public and private investments may lead to varying outcomes, while ensuring that each candidate site is served by no more than one entity. This scenario reflects a realistic policy context in which both government agencies and private businesses seek to develop EV chargers but must coordinate their siting decisions.

Let F^P and F^b denote the portions of total funding F allocated to public-led and private-led strategies, respectively, with the condition of Eq. (33).

$$F^P + F^b \leq F \tag{33}$$

Each actor then independently solves its own CSLP, based on its distinct objective function and decision-making logic. However, a critical shared constraint is that each candidate site $i \in I$ can be selected by at most one entity, ensuring no duplication of infrastructure deployment. We define this exclusivity constraint using the binary decision variables:

- $x_i^p = 1$: if a public-led charger is built at location i, 0 otherwise;
- $x_i^b = 1$: if a private-led charger is built at location i, 0 otherwise.

The mutual exclusivity constraint is then given by Eq. (34).

$$x_i^p \cdot x_i^b \leq 1 \tag{34}$$

The objectives of public and private CLSP problems are consistent to the previous sections. The objective of public sector is to maximize aggregate standardized occupancy across all selected public sites (Eq.(6), whereas the objective of private sector is to maximize aggregate profit across all selected private sites (Eq.(27).

Each CSLP problem is subject to its own constraints: Public sector funding constraint is formulated as Eq.(15). Private sector has three constraints, budget feasibility (Eq. (22)), business feasibility (Eq. (23)) and subsidy constraint (Eq. (32)).

In the public-private coexist scenario, public and private CSLP models are first solved independently under their respective objectives. When both models identify the same site as optimal, a siting conflict arises. To resolve this, we introduce a rule-based arbitration mechanism based on scaled benefit weights. Specifically, the public and private objective metrics are normalized onto a common scale using z-scores (Eq. (35))

$$z_i^p = \frac{\rho_i - \mu_\rho}{\sigma_\rho} \qquad z_i^b = \frac{\Pi_i - \mu_\Pi}{\sigma_\Pi} \tag{35}$$

where ρ_i is the predicted occupancy, Π_i is the predicted profit, and μ, σ represent the mean and standard deviation of their respective distributions. This standardization allows fair comparisons across differing units.

Each standardized score reflects the relative contribution of a site compared to all other candidates. When a conflict arises, the site is assigned to the actor for which the standardized score is higher, provided the budget constraint is satisfied. This rule is expressed as Eq. (36).

$$\begin{cases} x_i^p = 1,\ x_i^b = 0;\ \text{if } z_i^p > z_i^b \\ x_i^b = 1,\ x_i^p = 0;\ \text{if } z_i^p \leq z_i^b \\ x_i^p = 0,\ x_i^b = 0;\ \text{otherwise (infeasible or low priority site)} \end{cases} \tag{36}$$

This comparison-based arbitration mechanism is enforced during the iterative solution process. It ensures that only one actor receives the right to develop each contested site and that assignment reflects the entity that can generate higher marginal benefit from the deployment. This rule effectively serves as an additional constraint layered on top of each actor's individual CSLP. Similar priority-based and rule-driven coordination mechanisms have been widely adopted in facility location and multi-actor planning problems as practical tools for reconciling competing objectives over shared candidate sites (Drezner & Hamacher, 2004; Klose & Drexl, 2005). By embedding this conflict resolution mechanism, the model supports coordinated planning under budgetary limits and heterogeneous investment objectives.

3.2.4. Summary of CSLP model formulations

Table 2 summarizes the key modeling components of the three CSLP formulations under public-led, private-led, and coexistence scenarios.

Table 2. Comparison of the three CSLP model formulations.

Components	Public-led	Private-led	Coexistence
Decision variable	x_i^p, public siting decisions, Eqs. (1)-(2)	x_i^b, private siting decisions, Eqs. (16)-(17)	(x_i^p, x_i^b), public and private siting decisions, Eqs. (1)-(2) and (16)-(17)
Objective function	Maximize occupancy, Eq. (6)	Maximize profit, Eq. (27)	Public and private objectives, Eqs. (6) and (27)
Revenue components	Charging-related revenue, Eq. (7)	Charging-related and business revenue, Eq. (18)	Public and private revenue components
Cost components	Construction and operating costs, Eq. (10)	Construction and operating costs, Eq. (21)	Public and private cost components
Constraints	Public funding constraint, Eq. (15)	Budget, business, and subsidy constraints, Eqs. (22), (23), and (32)	Public and private constraints plus funding allocation and mutual exclusivity, Eqs. (33) and (34)
Secondary metric	Profit Π^p, Eq. (14)	Occupancy Φ^b, Eq. (31)	Public and private secondary metrics
Arbitration mechanism	Not applicable	Not applicable	Standardized occupancy and profit scores, Eqs. (35)-(36)

3.3. Lower-level ensemble learning model for charging demand prediction

The lower-level model estimates station-level charging utilization as a function of candidate site characteristics and the infrastructure layout passed from the upper-level model. This component is needed because new charger deployment changes the surrounding supply environment, which may in turn affect the expected occupancy of both new and existing stations. To capture this demand-supply feedback, we adopt a boosting-based machine learning approach that can learn nonlinear relationships between station utilization and contextual features, such as land use, station density, weather conditions, charger type, and nearby traffic activity. This data-driven approach is well suited to the available dataset, which contains long-term station-level charging records and multiple spatial and operational attributes. This approach also enables the framework to learn from aggregated charging behavior observed across stations and improve station-level utilization prediction. The predicted occupancy forms the basis for occupancy and profitability calculations, which are then used to update siting decisions iteratively.

3.3.1. Data overview

To support demand forecasting, we develop a data-driven model utilizing a multi-source dataset collected from Kansas City, Missouri (KCMO). For comprehensive details on the data sources and preprocessing procedures, please refer to Song and Hu (2023). The dataset integrates diverse information streams, each contributing unique contextual and behavioral insights relevant to EV charging demand. Specifically, the following components were collected:

- **Charging Event Logs (sourced from the ChargePoint cloud platform between 2017 and 2023):**

These logs include detailed information on individual charging sessions, such as session start and end times, initial and final state of charge (SOC), energy consumed, station ID, plug type, and geolocation. From this data, we construct the primary response variable—monthly occupancy at each charger ($\rho_{i,m}$)—along with key station-level features such as charger type and surrounding spatial characteristics.

- **Travel Demand Model Data (provided by the Mid-America Regional Council):** This dataset includes zonal trip productions and traffic volumes on adjacent roadways, offering proxy indicators for EV mobility patterns and potential charging demand across different geographic areas.
- **Climate Data (from the National Centers for Environmental Information):** Weather variables are incorporated to account for environmental factors that may influence charging behavior, such as seasonal variability in EV usage or range anxiety under adverse weather conditions.

3.3.2. Feature definitions and derivation from the raw dataset

To enable accurate modeling of charging demand, we define and extract a set of features that are potentially influential to charging behavior, drawing from the multi-source raw dataset described earlier. Specifically, we construct a feature vector, denoted as $a_{i,m}$, for each candidate location i and month m, where each element in the vector corresponds to a distinct location-level attribute that may affect charging usage. These features are categorized into the following groups:

First, we include charger contextual variables, beginning with a station density metric denoted as D_i, which reflects the *density of chargers* within the ZIP code area where station i is located. This variable, adapted from Song and Hu (2023), is intended to capture both the local intensity of infrastructure supply and the potential competitive effects among nearby stations. The station density is computed as Eq. (37).

$$D_i = N_i / A_i \tag{37}$$

where N_i is the total number of chargers in the same ZIP code area as station i, and A_i represents the geographic area (in square miles or kilometers) of that ZIP code. The value of N_i is determined using the Eq. (38).

$$N_i = \sum_{j \in I} x_j \cdot y_j^i \tag{38}$$

where x_j is the binary decision variable indicating whether site j is selected, and y_j^i is a spatial indicator defined as Eq. (39).

$$y_j^i = \begin{cases} 1 & \text{if } i \text{ and } j \text{ are in the same area} \\ 0 & \text{otherwise} \end{cases} \tag{39}$$

Incorporating D_i into the feature vector allows the model to account for the local charging supply environment. A higher charger density indicates more nearby charging options, which may reduce the expected use of each individual station if local demand is already well served. However, high density alone does not imply low utilization. The model learns the influence of D_i jointly with other contextual features, and the overall relationship is inferred from the data.

Next, we incorporate *temporal information*, denoted as $T_{i,m}$, defined as an integer index representing the temporal dimension. In this study, the lower-level model adopts monthly granularity to capture short-term variations in charging behavior. This temporal resolution allows the model to account for seasonal trends and behavioral shifts over time. By including $T_{i,m}$, the model becomes more responsive to temporal demand dynamics, enhancing its ability to forecast peak usage periods and support more precise infrastructure planning.

Meanwhile, weather variables, including *monthly averages of temperature, precipitation, wind speed and snowfall*, are incorporated, denoted as $TMP_{i,m}$, $PRCP_{i,m}$, $WD_{i,m}$ and $SF_{i,m}$ respectively. These monthly weather fluctuations significantly influence EV charging behavior by affecting both driver mobility and charging patterns. For example, colder temperatures and snowfall may reduce overall travel distances or increase energy consumption per trip, leading to changes in charging frequency and duration. Precipitation and wind conditions can similarly impact driver's willingness to use public charging infrastructure. By integrating these weather variables, the model accounts for environmental factors that cause temporal variability in demand, thereby enhancing its accuracy in predicting charging needs across different seasons and weather conditions.

We then examine location characteristics, represented by a set of binary indicators $LT1_i$ to $LT7_i$, which capture the *land use types* at location i. These categories include commercial, industrial, institutional, recreational, residential, transportation-related, and vacant land uses. Each variable takes a value of 1 if the corresponding land use is present at the site, and 0 otherwise. Land use plays a significant role in shaping EV user presence and expected dwell times. For instance, commercial and institutional areas often experience higher daytime charging demand due to work-related or service-oriented trips, while residential zones typically show increased activity during evenings and weekends. Including these variables allows the model to account for the behavioral context associated with different land use patterns, improving its ability to estimate location-specific charging demand.

Further, we incorporate *charger hardware attributes*, represented by binary indicators $PT1_i$ and $PT2_i$, which denote the presence of DC fast chargers and Level 2 chargers, respectively, at location i. These features capture the technological configuration of each station and are critical in shaping user preferences and charging behavior. The availability of different charger types influences both the duration and frequency of charging sessions. DC fast chargers typically attract users seeking quick, high-turnover charging—often for en-route top-ups—while Level 2 chargers are more suitable for longer dwell times, such as those associated with work, shopping, or residential parking. Including these attributes allows the model to better align predicted usage patterns with station capabilities.

Finally, we include *traffic contextual variables*, specifically $AADT_{i,t}$, which represents the annual average daily traffic (AADT) volume near location i during time period t. This variable serves as a proxy for local traffic demand, derived from adjacent roadway segments. AADT captures two critical factors influencing EV charging behavior: human mobility and station visibility. Higher traffic volumes indicate greater movement of potential EV users in the vicinity, increasing the probability that a station will be noticed and utilized. By incorporating $AADT_{i,t}$, the model is better equipped to reflect the spatial dynamics of EV accessibility and forecast usage based on real-world traffic patterns.

3.3.3. Gradient-boosted decision tree-based ensemble learning model

The primary objective of our predictive modeling approach is to minimize the prediction error between observed and predicted charger occupancy. To achieve robust and accurate predictions, we develop and train an efficient gradient-boosted decision tree framework to iteratively construct an ensemble of regression trees, progressively improving accuracy by fitting residual errors from previous iterations.

Formally, let the prediction target (monthly occupancy) for station i in month m be represented as $\rho_{i,m}$. The ensemble model approximates this value through an additive combination of multiple regression trees f_k, each trained sequentially to minimize the residuals from previous stages. We define the model prediction $\rho_{i,m}$ for the feature vector $a_{i,m}$ as Eq. (40).

$$\rho_{i,m}(a_{i,m}) = \sum\nolimits_{k\in K} f_k(a_{i,m}) \tag{40}$$

where K denotes the total number of regression trees in the ensemble, and $a_{i,m}$ is a feature vector that includes the above defined features, and is given by Eq. (41).

$$a_{i,m} = [D_i,\ T_{i,m},\ TMP_{i,m},\ PRCP_{i,m},\ WD_{i,m},\ SF_{i,m},\ LT1_i, \ldots\ LT7_i,\ PT1_i,\ PT2_i,\ AADT_{i,t}] \tag{41}$$

To optimize the parameters and structure of these trees, we formulate the following objective function (Eq. (42)).

$$Min\ L(\Theta) = \sum\nolimits_{i\in I_e^p \cup I_e^b} \sum\nolimits_{m\in M} \left(\rho_{i,m}^{obs} - \rho_{i,m}\right)^2 + \sum\nolimits_{k\in K} \Omega(f_k) \tag{42}$$

Where $\Theta = \{f_1, f_2, \cdots, f_K\}$ represents the ensemble of regression trees, and $\Omega(f_k)$ denotes the regularization term to penalize model complexity and prevent overfitting. In the first term, $\rho_{i,m}^{obs} - \rho_{i,m}$ represents the prediction error, defined as the difference between the observed field data and the predicted values. M, here, represents the total number of months covered by the data.

The regularization term $\Omega(f_k)$ can be expressed as Eq. ((43)).

$$\Omega(f_k) = \gamma|l_k| + \frac{1}{2}\lambda||w_k||^2 \tag{43}$$

Where $|l_k|$ represents the number of leaves in the kth regression tree, and w_k represents for the vector of leaf weights. γ and λ are the hyperparameters controlling the strength of regularization, adjustable via hyperparameter tuning.

We employ a leaf-wise tree growth strategy combined with histogram-based binning to optimize training efficiency (Eq. (44)). During each iteration, the algorithm fits a new tree f_k to the residuals computed as the negative gradients of the loss function with respect to current predictions.

$$\varphi_{i,m}^{(k)} = -\frac{\partial L\left(\rho_{i,m}^{obs,k-1}, \rho_{i,m}^{(k-1)}\right)}{\partial \rho_{i,m}^{(k-1)}} \tag{44}$$

where $\rho_{i,m}^{(k-1)}$ denotes the cumulative prediction up to tree $k-1$. Each newly added tree then refines the

previous predictions (Eq. (45)).

$$\rho_{i,m}^{(k)} = \rho_{i,m}^{(k-1)} + f_k(a_{i,m}) \tag{45}$$

After completing the ensemble construction, the model generates monthly predictions of station occupancy. To derive annual occupancy predictions for each station location i in year t, we aggregate the monthly occupancy estimates within that year (Eq. (46)).

$$\rho'_{i,t} = \sum_{m} \rho_{i,m} \tag{46}$$

However, since the above equations forecast occupancy purely based on historical observations, it does not inherently account for future trends such as the anticipated growth in EV adoption. To incorporate such external macro-level insights, we introduce an additional calibration step that adjusts predictions according to an externally derived compound annual growth rate (CAGR) for EV adoption, denoted as ζ (Eq. (47)).

$$\rho_{i,t} = \rho'_{i,t}(1+\zeta)^t \tag{47}$$

The calibrated annual occupancy $\rho_{i,t}$ then serves as input into the upper-level optimization model (as specified in Eqs. (3), (4), (28) and (29)), enabling dynamic feedback and integration within the bi-level modeling framework. This integrated approach ensures that our station siting decisions remain responsive not only to immediate historical patterns but also to anticipated shifts in long-term EV adoption trajectories.

3.4. Solution algorithm with gradient-informed spatial expansion

To solve the proposed models, we develop a Gradient-Informed Spatial Expansion (GISE) algorithm that extends the iterative coupling scheme of Song and Hu (2023) with three modifications required by the multi-actor setting: (i) an explicit secondary-effect term that quantifies competitive draw from neighboring existing stations (Eq. (49)), which does not arise in single-planner solvers; (ii) an adaptive momentum factor that dynamically adjusts the exploration–exploitation balance (Eq. (52)); and (iii) a shared-site coordination procedure (Algorithm 2) for the coexistence scenario in which public and private deployments bid for the same candidate locations. The CSLP is formulated as a binary optimization problem, where decision variables (x_i^p, or x_i^b) take values of 0 or 1 to indicate whether a candidate site is selected.

To enhance computational tractability, we first apply linear relaxation to the binary decision variables. For instance, in the case of public chargers, the binary variable $x_i^p \in \{0,1\}$ is relaxed to a continuous variable $x_i^p \in [0,1]$. This relaxation allows the use of gradient-based optimization techniques in the early stages of the solution process, enabling efficient exploration of the solution space before reapplying integrality constraints in the final step.

We start with public chargers. At each iteration q, the algorithm evaluates the marginal contribution of developing a public charger at each candidate location $i \in I_c^p$. This is achieved by computing the first-order derivative of the objective function with respect to the relaxed siting variable $x_{i,q}^p$. As shown in Eq.(48), this derivative corresponds to the cumulative predicted output of the lower-level demand model over the

evaluation period.

$$\frac{d\left(x_{i,q}^{P}\rho_{i,q}^{p}\right)}{dx_{i,q}^{p}} = \sum_{t\in T}\rho_{i,t,q}^{p} = \sum\nolimits_{m\in M}\sum\nolimits_{k\in K} f_k(a_{i,m,q}^{p}) \tag{48}$$

This derivative serves as an approximation of the marginal benefit associated with placing a station at location i, capturing its contribution to the overall system objective (e.g., occupancy or profit). By quantifying the impact of incremental siting decisions, this gradient-based measure enables the algorithm to prioritize high-value candidate sites during each iteration. In doing so, the GISE algorithm efficiently navigates the solution space, incrementally expanding the spatial configuration based on data-driven utility signals from the lower-level model.

In addition to the direct benefits of deploying a new charger at location i, the algorithm also accounts for secondary impacts on nearby existing stations $j \in I_{e,i}^{p}$, located within the same ZIP code area (i.e., $y_i^j = 1$). These impacts arise due to spatial competition—as new stations may draw users away from nearby sites—and are quantified as the difference in cumulative output from the lower-level model before and after location i is selected.

This secondary effect is captured in Eq. (49):

$$\Delta\rho_{e,i,q}^{p} = \sum_{j\in I_{e,i}}\sum_{t\in T}(\rho_{j,t,q}^{p} - \rho_{j,t,q=0}^{p}) = \sum_{j\in I_{e,i}}\sum\nolimits_{m\in M}\sum\nolimits_{k\in K}(f_k(a_{j,m,q}^{p}) - f_k(a_{j,m,q=0}^{p})) \tag{49}$$

The total marginal contribution $\delta_{i,q}^{p}$ for siting a public charger at location i during iteration q is defined as the sum of both the direct effect (i.e., the incremental benefit from adding a station at i) and the indirect effect (i.e., the secondary impact on nearby existing stations). This comprehensive metric captures the net system-level benefit of selecting location i, accounting for both positive gains and potential competitive drawbacks. The formulation is given in Eq. (50):

$$\delta_{i,q}^{p} = d(x_{i,q}^{P}\rho_{i,q}^{p})/dx_{i,q}^{p} + \Delta\rho_{e,i,q}^{p} \tag{50}$$

This value $\delta_{i,q}^{p}$ serves as the key selection criterion in the GISE algorithm, guiding the spatial expansion process by identifying locations with the highest overall contribution to system performance. By jointly considering local potential and network-wide interactions, the algorithm supports more informed and coordinated siting decisions.

This process underscores the tight coupling between the upper and lower-levels of the bi-level framework. Each siting decision dynamically updates the input feature vector $a_{i,m,q}^{p}$, which feeds into the lower-level predictive model to revise occupancy forecasts. This feedback mechanism ensures that the siting decisions and demand estimates co-evolve iteratively. After each iteration, key contextual variables such as station density D_i are recalculated using Eqs. (37)-(39), allowing the lower-level model to reflect the most current charger deployment landscape.

A similar approach is applied for private charger siting. The total marginal contribution at candidate location

$i \in I_c^b$, denoted as $\delta_{i,q}^b$, integrates both the direct gain in profit and the secondary impacts on existing nearby private stations. This is expressed in Eq. (51):

$$\delta_{i,q}^b = d(x_{i,q}^b \Pi_{i,q}^b)/dx_{i,q}^b + \Delta\Pi_{e,i,q}^b \tag{51}$$

Where $d(x_{i,q}^b \Pi_{i,q}^b)/dx_{i,q}^b$ is the marginal profit generated by selecting location i, and $\Delta\Pi_{e,i,q}^b$ represents the induced change in profit at nearby private stations due to competitive interaction.

The GISE algorithm is designed to iteratively identify candidate locations with the highest marginal contribution $\delta_{i,q}$. To achieve a balance between local exploitation and global exploration, the search process evaluates two types of candidate sites: (1) adjacent neighbors N_i, which support fine-tuned local refinement, and (2) randomly sampled distant candidates $n_{i,q}$, which enable broader search and help avoid entrapment in local optima. This hybrid search strategy improves convergence speed and robustness of the solution process.

The number of distant candidates sampled in each iteration is dynamically adjusted using Eq. (52):

$$n_{i,q} = N_i^a \cdot n_0(1 + 0.1q_{\pm}) \tag{52}$$

Where N_i^a is the number of candidate locations not adjacent to i, $n_0 \in (0,1]$ is the initial sampling ratio, and $q_{\pm}$ is an adaptive momentum factor reflecting the cumulative number of iterations with or without successful improvements in $\delta_{i,q}$.

Specifically, when a new iteration results in identifying a location with higher marginal contribution, $q \pm$ is incremented positively; otherwise, it is decremented. If the trend switches (from success to failure or vice versa), the count is reset to zero and begins accumulating in the new direction. This adaptive mechanism enables the algorithm to dynamically adjust the search radius, becoming more exploratory when progress slows and more exploitative when convergence is strong.

The iterative coupling search algorithm proceeds through a feedback-driven process that integrates the lower- and upper-level models. The procedure consists of the following key steps:

1. Demand Prediction: The lower-level model generates station-specific occupancy predictions based on the current feature vectors.
2. Site Selection: The upper-level model evaluates all candidate sites—both adjacent and randomly sampled distant locations—and selects the site i with the highest total marginal contribution $\delta_{i,q}$.
3. Feature Update and Feedback: The siting decision updates the corresponding feature vectors $a_{i,m,q}^p$, which in turn retriggers the lower-level model to recompute occupancy forecasts, incorporating the latest deployment status.

This feedback loop continues iteratively until convergence is achieved, either when no further improvement in marginal contribution (e.g., $|\sum_i \delta_{i,q} - \sum_i \delta_{i,q-1}| < \kappa$) is found or a pre-defined stopping criterion (e.g., iteration limit or budget exhaustion) is met.

To further clarify the implementation details, the following pseudocode outlines the algorithmic framework. We first present the structure for separate investment scenarios (i.e., public-only or private-only) in Algorithm 1, followed by the framework for the public–private coexistence scenario in Algorithm 2, which incorporates inter-model coordination and shared-site constraints.

Algorithm 1: GISE for public- or private-led scenario

Input:

Existing location set I_e^p or I_e^b; Candidate location set I_c^p or I_c^b

Fund $F^p = F$ or $F^b = F$

Investment period T; Discount rate r; Initial selection ratio n_0

Lower-level model, feature array $a_{i,m}$ and calibration coefficient ζ

Initialization:

Solution set $Sol = \emptyset$

Repeat: Randomly select $i \in I_c^p$ or I_c^b, solution set $Sol \leftarrow Sol \cup i$, until constraint is not satisfied

Solution procedure:

Repeat until Sol across successive iterations stay unchanged

For each $i \in Sol$, do

Identify N_i and $n_{i,q}$ candidate locations called j, compute their marginal contribution $\delta_{j,q}$

Select the maximum value among $\delta_{i,q}$ and $\delta_{j,q}$

If constraint is satisfied:

Add the j^* with maximum value to $Sol \leftarrow Sol \cup j^*$

Set $i \leftarrow j^*$

Update $a_{i,m}$ and invoke lower-level model ($a_{i,m}$)

Else if there exists several $i \in Sol$ with cumulative $\delta_{i,q} < \delta_{j^*,q}$ and constraint is satisfied

Remove i from solution: $Sol \leftarrow Sol - i$

$Sol \leftarrow Sol \cup j^*$ and $i \leftarrow j^*$

Update $a_{i,m}$ and invoke lower-level model ($a_{i,m}$)

Else

Skip this j^* and continue

Output: Solution set Sol

Algorithm 2: GISE for coexistence scenario

Input: Same as Algorithm 1

Initialization: Same as Algorithm 1

Solution procedure:

Run Algorithm 1 with given F^p and F^b, respectively

Repeat until Sol^p and Sol^b across successive iterations stay unchanged

For each $i \in I_c^p \cap I_c^b$, do

If $z_i^p > z_i^b$

$I_c^p \leftarrow I_c^p \cup i$

Else

$I_c^b \leftarrow I_c^b \cup i$

Invoke Algorithm 1 with updated I_c^p and I_c^b, respectively.

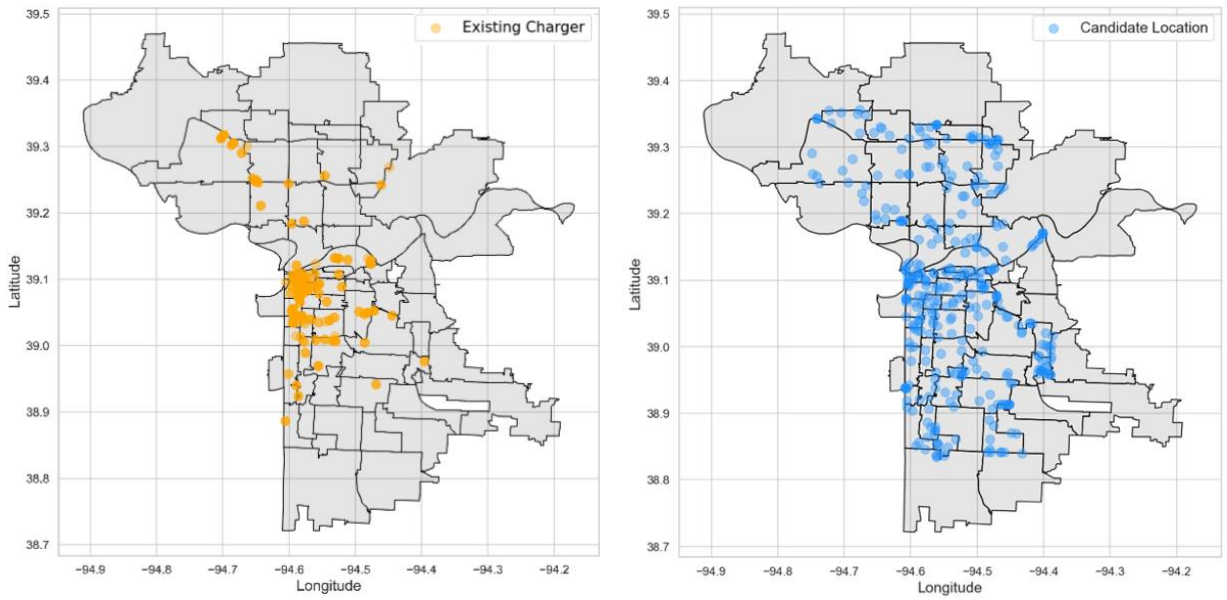
Output: Solution sets Sol^p and Sol^b

4. Case study

This section presents a real-world case study designed to evaluate the employed bilevel optimization framework and GISE algorithm. Using data from Kansas City, Missouri (KCMO), we illustrate the applicability of the framework to large-scale EV charger planning. Kansas City was selected because it provides long-term station-level charging records and supporting spatial, traffic, climate, and business-location data required to implement the proposed framework. The case study is intended as an empirical demonstration rather than a nationally representative sample. As a mid-sized, car-oriented U.S. metropolitan area with mid-stage EV adoption, Kansas City provides a suitable context for evaluating public-led, private-led, and coexistence deployment strategies. The quantitative results should therefore be interpreted in light of local charging behavior, electricity prices, travel patterns, and business composition. Nevertheless, the case study offers broader policy-relevant insights regarding the leverage effect of public funds, the user-cost implications of private charging fees, and the dependence of private profitability on charging-to-purchase conversion behavior. The framework itself can be recalibrated for other cities with comparable data availability. The results include the performance of the lower-level station-level utilization prediction model and the upper-level deployment strategies under three investment scenarios, followed by discussion of practical policy implications.

4.1. Data Description and Parameter Settings

The dataset used in this case study comprises 416,647 charging event records collected from 435 existing chargers between 2017 and 2023. All 435 existing chargers in the dataset are public chargers. The dataset also includes annual average daily traffic (AADT) data from the Mid-America Regional Council (MARC) and climate data from the National Centers for Environmental Information (NCEI). In addition, 355 candidate locations are identified for new public or private chargers, with each candidate site associated with an existing business. Figure 3 illustrates the spatial distribution of existing chargers (orange dots) and candidate locations (blue dots) within the city boundary (gray polygons). Existing chargers are mainly concentrated in urban centers, whereas candidate locations are distributed relatively evenly. Based on the dataset, Level 2 chargers account for 98.6% of all existing chargers, while DC fast chargers represent only 1.4% of the total. Of the Level 2 chargers, 99.8% are equipped with two ports, whereas only 0.2% have a single port. These statistics demonstrate that nearly all existing public charging infrastructure in the study area consists of dual-port Level 2 chargers.



(a) (b)

Figure 3. Spatial distribution of the (a) existing chargers and (b) candidate locations

Figure 4 presents histograms of key operational metrics across the dataset of existing chargers. The top-left subfigure shows the monthly occupancy distribution, with most stations recording fewer than 100 hours per month. The top-right subfigure illustrates overstay parking duration per month, which follows a similar highly skewed pattern. The bottom-left subfigure displays energy consumption per month, showing that most stations consume less than 500 kWh monthly. The bottom-right subfigure presents the number of charging sessions per month, with the majority of stations serving fewer than 50 sessions. Notably, the distributions of overstay parking duration, energy consumption, and charging sessions closely follow the highly skewed pattern of station occupancy, with long tails indicating substantial variation in station-level usage. These observations support the use of occupancy as a primary predictor for estimating other charging metrics in the upper-level model.

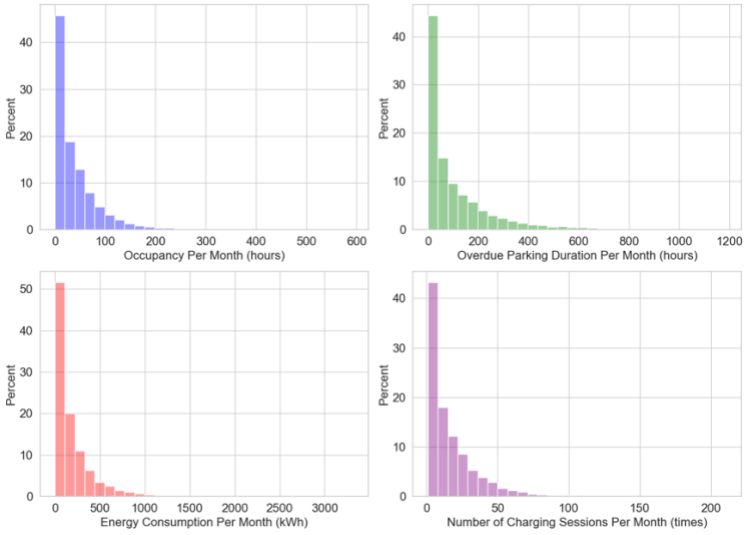


Figure 4. Distribution of charging metrics in existing chargers

All critical parameters of the model used in the case study are summarized in Table 3:

Table 3 Key input parameter settings for case study simulations

Parameter	Value	Description / Source
T	5 years	Planning horizon/investment period
F	\$500,000	Maximum allowable fundings
r	0.03	Discount rate (Edenhofer, 2015; Stern, 2007)
N_e^P, N_e^b	435, 0	Number of existing public and private chargers
N_c^P, N_c^b	355, 355	Number of public and private candidate locations
$s_{i,t}^p, s_{i,t}^b$	\$0, \$0.1/kWh	Service fee for public and private chargers (Wood et al., 2023)
$\varepsilon_{i,t}^p, \varepsilon_{i,t}^b$	8.55¢/kWh, \$0.21/kWh	User charging fee, public set at commercial electricity rate and private based on observed KCMO rate
$u_{i,t}^p, u_{i,t}^b$	\$0, \$1/hour	Overstay parking fee for public and private chargers (Wood et al., 2023)
$\varepsilon_{i,t}$	8.55¢/kWh	KCMO commercial electricity rate
C_i^p, C_i^b	DC fast: \$60k; Level 2: \$15k	Construction cost (MoreDay, 2024; Smith & Castellano, 2015)
$\omega_{i,t}^p, \omega_{i,t}^b$	0.1	Annual additional operating costs-to-construction cost ratio (Smith & Castellano, 2015)
$\pi_{i,t}$	Fast-food restaurant: \$1–\$5; Service station/convenience store: \$5–\$10; Supermarket: \$10–\$20; Shopping center: \$20-\$40; Hotel: \$40-\$50	Median profit per transaction; for each candidate, drawn independently and uniformly within the interval corresponding to its associated business type, so same-type candidates differ. (NACSStores 2023, IBISWorld 2024, NRAAssociation 2025, IBISWorld 2025, IBISWorld 2025)
$\alpha_{i,t}$	0.1	Probability of in-store purchase (scenario assumption, examined in Section 4.5.4)
$\beta_{i,t}$	0.95	Discount coefficient
B_i	0.2	Private budget-to-construction cost ratio (USDOT, 2025)
ζ	0.12	CAGR for EV adoption (Wood et al., 2023)
κ	0.01	Convergence threshold

4.2. Evaluation of Lower-Level Prediction Model Accuracy

In this subsection, we evaluate the accuracy of the developed ensemble learning model at the lower-level and compare its performance against several benchmark models. Using the provided dataset, 80% was allocated for model training and 20% for testing. After training, a total of 24,724 sub-regression trees were constructed. Model performance on both training and testing datasets was evaluated using R^2, root mean squared error (RMSE) and mean absolute error (MAE).

As illustrated in Figure 5, the training subset achieved an R^2 of 0.79, RMSE of 22.57, and MAE of 15.48, whereas the corresponding testing dataset results were 0.70, 27.68, and 18.07, respectively.

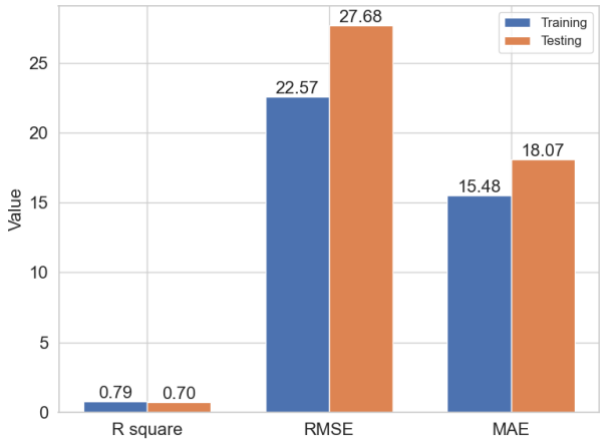


Figure 5. Prediction performance for occupancy on training and testing sets

To further validate the model's predictive performance, four additional benchmark models were applied to the same dataset: Multiple Linear Regression (MLR), Random Forest Regression (RF), K-Nearest Neighbors (KNN) and Deep Neural Network (DNN). As summarized in Table 4, the proposed lower-level model achieved the best performance among the tested models indicating relatively high predictive accuracy.

Table 4 Performance comparison among five models

Model	R^2	RMSE	MAE
Proposed model	0.70	27.68	18.07
MLR	0.05	48.91	32.85
RF	0.68	28.37	17.65
KNN	0.55	33.72	21.49
DNN	0.47	36.70	23.86

Moreover, we conducted descriptive statistical analysis of both observed and predicted values, as presented in Figure 6. The predicted values closely align with the observed values in terms of mean, median, maximum, and standard deviation, demonstrating strong predictive accuracy. The only notable deviation occurred for minimum values, where some existing chargers exhibited extremely low occupancy, leading to predicted values near zero or slightly negative.

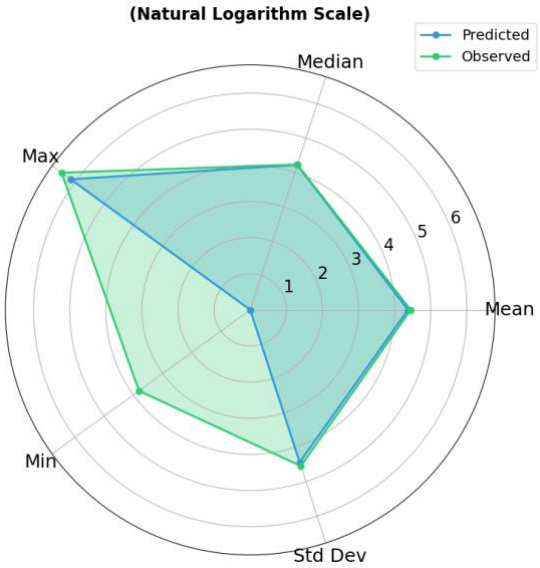


Figure 6. Comparison of the statistical measures of observed and predicted values

To assess feature importance, the Shapley Additive Explanations (SHAP) method was employed, which assigns consistent contribution values by averaging over all possible permutations of feature order. Figure 7 presents the top ten most important features, ranked by mean SHAP value. The most influential predictor was nearby traffic volume, measured by AADT. It is worth noting that the density of neighboring chargers ranked third in importance, underscoring the value of explicitly modeling the demand–supply coupling between the upper- and lower-level components of the framework.

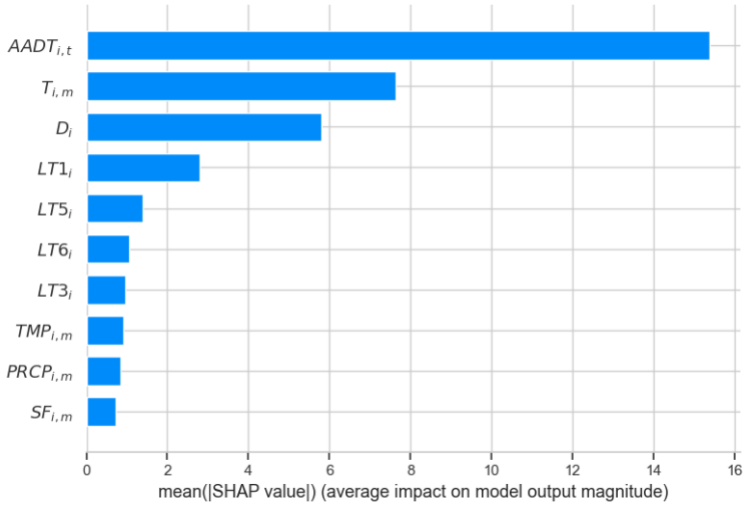


Figure 7. The mean SHAP value for the 10 most important features

4.3. Algorithm performance and convergence

We solve the developed bi-level modeling framework using the proposed GISE solution algorithm, and the resulting performance comparisons are presented below. We first conduct experiments under the public-led

scenario to examine the convergence behavior of GISE algorithm.

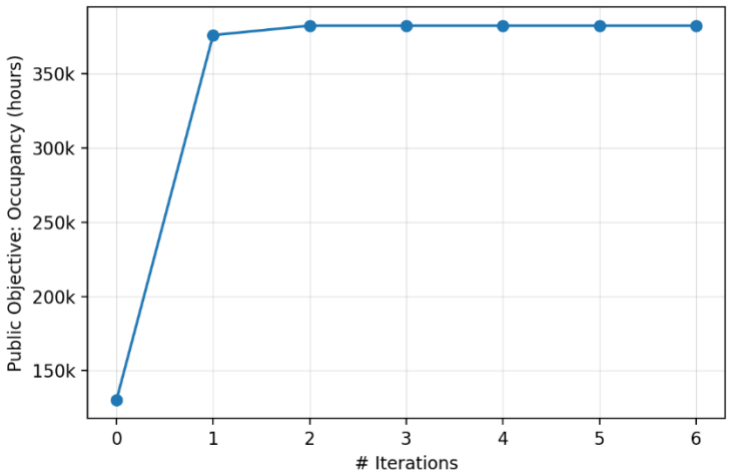


Figure 8. Convergence of GISE algorithm in the public-led scenario

The convergence of GISE algorithm is illustrated in Figure 8, where the x-axis represents the number of iterations and the y-axis corresponds to the public objective function value measured by occupancy (hours). As the iteration progresses, the occupancy increases rapidly in the initial stage, followed by a slower growth rate, and eventually stabilizes. Specifically, the occupancy improves from the initial solution of 130,354 hours to the final optimal solution of 382,464 hours after six iterations. Moreover, the occupancy exhibits only marginal changes after the third iteration, indicating that GISE algorithm demonstrates fast and stable convergence.

To validate the performance of GISE, we compare it to (1) a Greedy-based Algorithm (GA), which directly selects the initially optimal location. We refer to this approach as "greedy-based" because it selects the highest-ranked sites independently, without considering the demand-supply coupling relationship between the upper- and lower-levels; and (2) a Neighborhood Search Algorithm (NSA), which searches only in the local neighborhood for a better solution.

Figure 9 illustrates the performance of these algorithms under the same investment period and funding level. The figure reports both occupancy (left y-axis) and profit (right y-axis), capturing outcomes for both public and private objectives, as well as their secondary metric (public profit and private occupancy). It can be seen that in the public charger scenario—where the entire $500,000 fund was allocated to building public chargers—the GISE algorithm achieved the highest occupancy (blue bars) at 382,464 hours. This represents a substantial improvement over the initial solution at 130,354 hours (an increase of 193.4%), the GA solution at 373,531 hours (2.4% improvement), and the NSA solution at 377,680 hours (1.3% improvement). As for the secondary metric, profit, all algorithms yield identical results. This is because, under the assumptions of this case, public chargers only charge users a fee equal to the electricity cost.

Figure 9 also shows that in the private charger scenario—where the entire $500,000 fund was allocated to building private chargers—the GISE algorithm achieved the highest profit (red bars) of $692,000, representing a significant improvement over the initial solution at $344,888 (an increase of 100.6%), the

GA solution at $641,040 (7.9% improvement), and the NSA solution at $639,626 (8.2% improvement). Interestingly, although occupancy was not explicitly optimized, the GISE algorithm also produced the highest occupancy (green bars) at 544,535 hours. In comparison, the occupancies for the initial solution, GA solution, and NSA solution were 341,213, 498,075, and 496,982 hours, respectively—marking improvements of 59.6%, 9.3%, and 9.6%. These results suggest that, in a private-led scenario, the GISE algorithm not only maximizes profit but also achieves the highest occupancy as a beneficial side effect, outperforming both the initial solution and the benchmark algorithms.

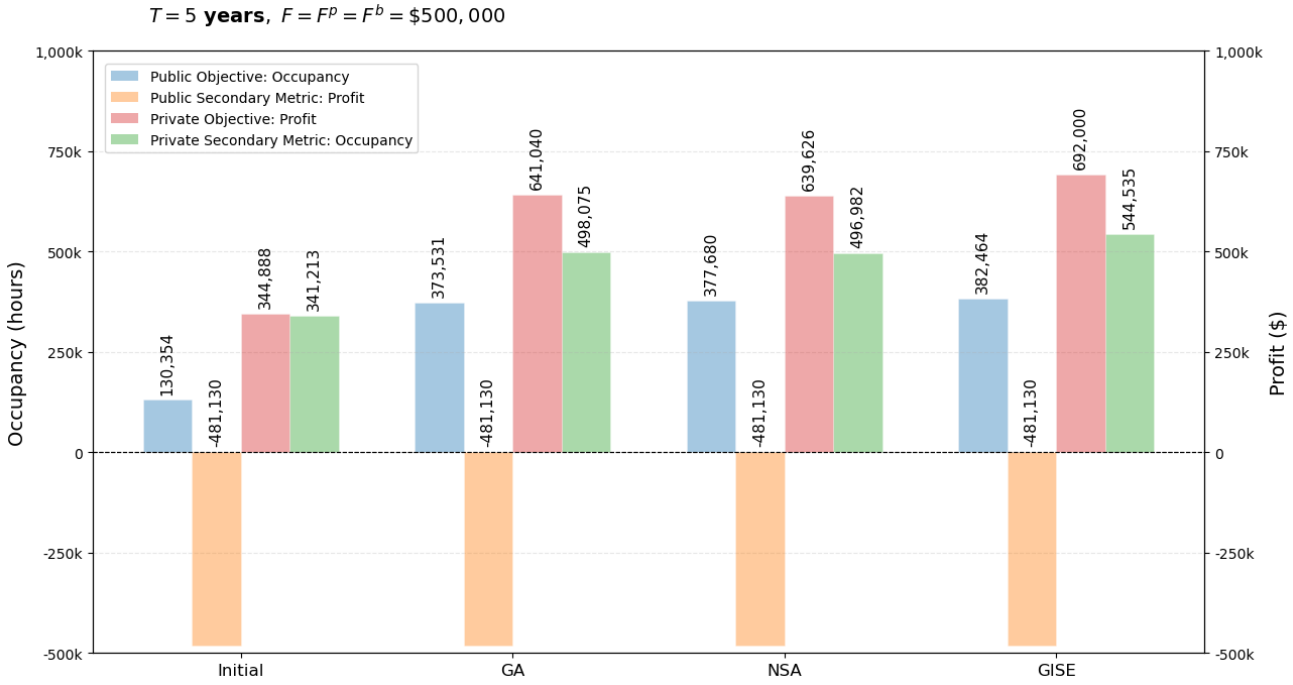


Figure 9. Performance comparison of different algorithms

4.4 Public-private deployment outcomes and policy trade-offs

To investigate how limited government funding can be allocated between public-led and private-led deployment, we conducted experiments under the coexistence scenario. Figure 10 presents the results over a 5-year period, where a total of $500,000 is allocated between public and private chargers in varying proportions. The allocation begins with $0 directed to private chargers (i.e., $F^b = 0$, F^p=$500,000) and gradually increases F^b to $500,000 in increments of $100,000. The bar chart reports the occupancy and profit of public chargers, shown by the blue and orange bars, and private chargers, shown by the green and red bars. Additionally, three dashed lines depict the total profit (red dashed line), total occupancy (blue dashed line) and additional charging costs (green dashed line) across all chargers. Additional charging costs refer to the extra fees users pay when chargers are operated by private investors. These costs include service fee, overstay parking fee and the portion exceeding the commercial electricity rate. They do not include optional in-store purchases. When all funding is allocated to public chargers, no additional charging costs are incurred. As F^b increases, additional charging costs gradually rises.

The results show that allocating a greater share of public funding to private-led deployment reduces public-charger occupancy but substantially increases private-charger occupancy. A notable shift occurs when F^b increases from 0 to \$100,000, marking the transition from a public-only scenario to a coexistence scenario, where public-charger occupancy drops sharply, while private-charger occupancy increases. This suggests that those candidate locations that can contribute more occupancy have larger private z-scores z_i^b than public z-scores z_i^p, and are thus preferred to build private chargers. The profit of public chargers decreases linearly because each public charger has a fixed profit. In contrast, the profit growth for private chargers shows a decelerating trend, while their occupancy grows steadily. This indicates that occupancy and profit are not fully correlated: while some candidates achieve both high occupancy and high profit, there also exist candidates with high profit but low occupancy, and vice versa.

When both charger types are considered together, allocating more public funding to private-led deployment is associated with higher total occupancy and total profit. This reflects the leverage role of public subsidies, which increases the deployment efficiency of public funds by mobilizing additional private capital. In the private-led case, public funding is combined with required private capital contributions, enabling a larger number of chargers to be deployed for a given level of government expenditure. For example, as shown in Figure 10, a public budget of \$500,000 can support the deployment of 22 public stations when fully allocated to public-led investment. In contrast, when fully allocated to private-led deployment, the same \$500,000 in public funds can support up to 41 private stations, together with approximately \$125,000 in private co-investment under the assumed 20% capital contribution requirement.

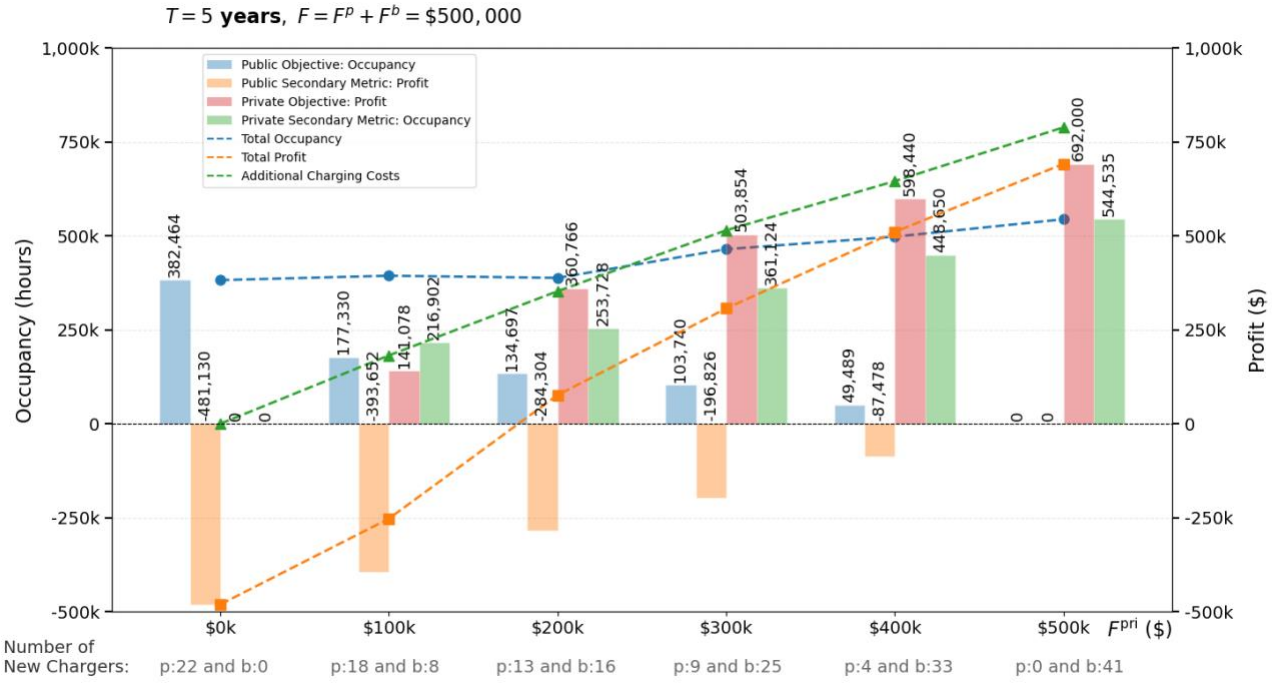


Figure 10. Results across different funding allocations

Notably, total occupancy and total profit do not increase at the same rate. While total occupancy increases modestly, total profit increases more substantially as a larger share of public funding is allocated to private

chargers. This trend suggests that private stations possess greater revenue-generating capacity, likely due to operational and business efficiencies supported by private investment. However, it also increases additional user costs, as residents face higher prices for charging and parking. Quantitatively, private-led deployment imposes an additional charging cost of approximately $1.40 per occupancy hour relative to public-led deployment. For frequent public-access charging users in the Kansas City dataset, with 4–10 charging events per month and an average session duration of 2.88 hours, this equals approximately $4.03 per event, or $193-$484 per year. Relative to the $800-$1,300 annual fuel-cost savings that EV owners typically obtain over comparable gasoline vehicles (Harto, 2020), this additional expense offsets roughly 24%-37% of that benefit. This suggests that private-led deployment may reduce the cost-saving advantage of EV adoption for users dependent on public-access charging. From a policy perspective, private-led deployment therefore involves a trade-off between infrastructure expansion and user affordability. A larger number of private chargers may improve charging accessibility under constrained public budgets, but it may also shift part of the infrastructure efficiency gain onto frequent public-access charging users through higher costs. Subsidy programs for private chargers may therefore need affordability safeguards, such as fee caps or targeted discounts, to balance expanded access with user cost protection.

A further dimension of the public–private trade-off is spatial coverage. Figure 11 compares the siting solutions under the public-led and private-led scenarios at the $500,000 budget level. The overlap is 84.2% at the postal-code level, with 16 of 19 public-led postal codes also served under private-led deployment, and 63.6% at the candidate-site level, with 14 of 22 public-led sites also selected under private-led deployment. Private-led deployment additionally extends coverage to 12 postal codes not reached by public-led deployment and establishes denser service in high-demand areas, with two or more stations in 11 postal codes under private-led deployment, compared with 3 postal codes under public-led deployment. Table 5 reports how the postal codes served by each solution are distributed across income quartiles, where quartiles are defined by the median household income (U.S. Census Bureau, 2024). In absolute terms, private-led deployment serves a similar number of low-income (Q1) postal codes as public-led deployment (five versus six), but serves more in other categories, giving it a larger total overall (28 versus 19). These results indicate that private-led deployment can expand aggregate service coverage by leveraging private matching capital.

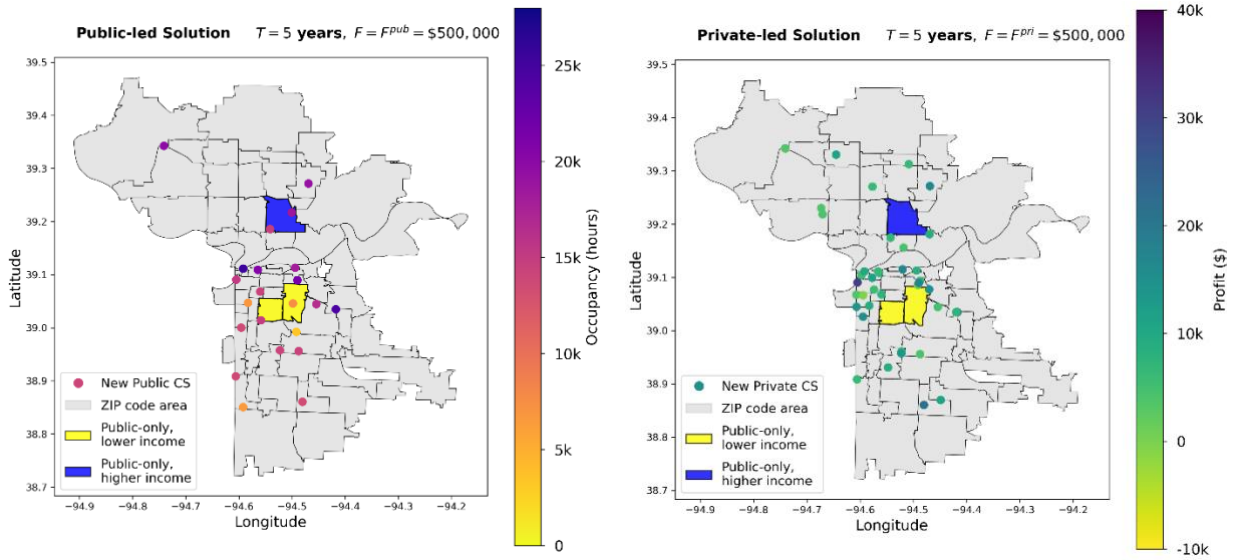


Figure 11. Spatial distribution of the (a) public-led solution and (b) private-led solution

Table 5 Distribution of postal codes served across income quartiles under the public-led and private-led solutions

Income quartile (median household income)	Public-led, n (%)	Private-led, n (%)	Postal codes served by both solutions
Q1 (< $53,888)	6 (31.6%)	5 (17.9%)	4
Q2 ($53,888–$66,036)	3 (15.8%)	6 (21.4%)	3
Q3 ($66,036–$98,316)	5 (26.3%)	6 (21.4%)	4
Q4 (≥ $98,316)	3 (15.8%)	7 (25.0%)	3
No residential income data	2 (10.5%)	4 (14.3%)	2
Total	19 (100%)	28 (100%)	16

Note: n is the number of postal codes served in each quartile; % is that quartile's share of all postal codes served by the solution.

This broader coverage, however, is distributed unevenly. Low-income (Q1) postal codes account for only 17.9% of the postal codes served by private-led deployment, compared with 31.6% under public-led deployment. Private-led deployment therefore tends to skew toward higher-income areas as its overall footprint grows. Specifically, as shown in Figure 11, three highlighted areas selected under public-led deployment are not selected under private-led deployment. Although these areas have relatively high absolute occupancy, their marginal profit contribution is outcompeted by other candidate sites under the private profit objective. Based on U.S. Census Bureau 2020–2024 ACS 5-year estimates (U.S. Census Bureau, 2024), the two yellow areas are predominantly African American communities with median household incomes of $51,310 and $40,770, both below the Kansas City median of $67,449, while the blue area is a higher-income predominantly White community with a median household income of $83,929.

Because the unmatched areas span both low- and higher-income quartiles (Q1 and Q3), this pattern is not a uniform withdrawal from low-income minority neighborhoods. However, two of the three are low-income minority communities, which together with the lower relative coverage in Table 5, points to a distributional concern that merits attention: profit-driven deployment may expand aggregate coverage while still bypassing some lower-income minority areas that public-led planning would serve. For policy design, this observation suggests that even when a private-led strategy is selected for aggregate efficiency, targeted interventions, such as additional incentives or set-aside funds for underserved communities, may be warranted to address demographic imbalances that a profit-driven outcome does not resolve.

This policy discussion, however, should be read together with a limitation of the demand model. Given the available data, the lower-level prediction model is trained on historical usage at public chargers and does not include price as a feature, so it does not capture how charging demand would respond to various prices. Because private stations generally adopt higher prices, charging demand at these stations would tend to fall, so the predicted private-station occupancy, and the profit derived from it, would be overestimated. The private-led advantages reported here, which depend on predicted utilization, should therefore be interpreted as upper estimates.

4.5. Sensitivity analysis

To further examine the trends in profit and occupancy for both public and private chargers, we conducted sensitivity analyses across varying funding levels and planning horizons.

4.5.1. Impact of varying funding levels

In this analysis, we examine how different levels of total funding influence the performance of public and private chargers over a 5-year period. The total funding F varies from \$200,000 to \$1,000,000 in \$200,000 increments. Two scenarios are considered: one where all funding is allocated to public chargers, and another where all funding is allocated to private chargers.

As in previous sections, Figure 12 uses blue and orange bars to represent occupancy and profit in the public-led scenario, and green and red bars for the private-led scenario. Lines are also used to illustrate trends across funding levels for easier comparison.

- Private charger Performance: The profit from private chargers (red bars) increases at a gradually slowing rate as funding grows. This suggests a saturation effect—likely due to the limited number of high-quality merchant partners or optimal locations. Additional funding may result in deploying chargers at lower-quality sites, leading to diminishing returns. The occupancy (green bars) follows a similar upward trend.

- Public charger Performance: Public charger occupancy (blue bars) increases almost linearly with funding, indicating steady growth in usage capacity. This trend suggests that public charger deployment is less impacted by the declining quality of later-added sites. One possible reason is that building public chargers can only use fixed government funding and cannot attract private capital, resulting in fewer units being deployed at each funding level. As a result, public chargers are less affected by the saturation effect compared to private chargers. In contrast, the profit of public chargers (orange bars) exhibits a consistent downward trend.

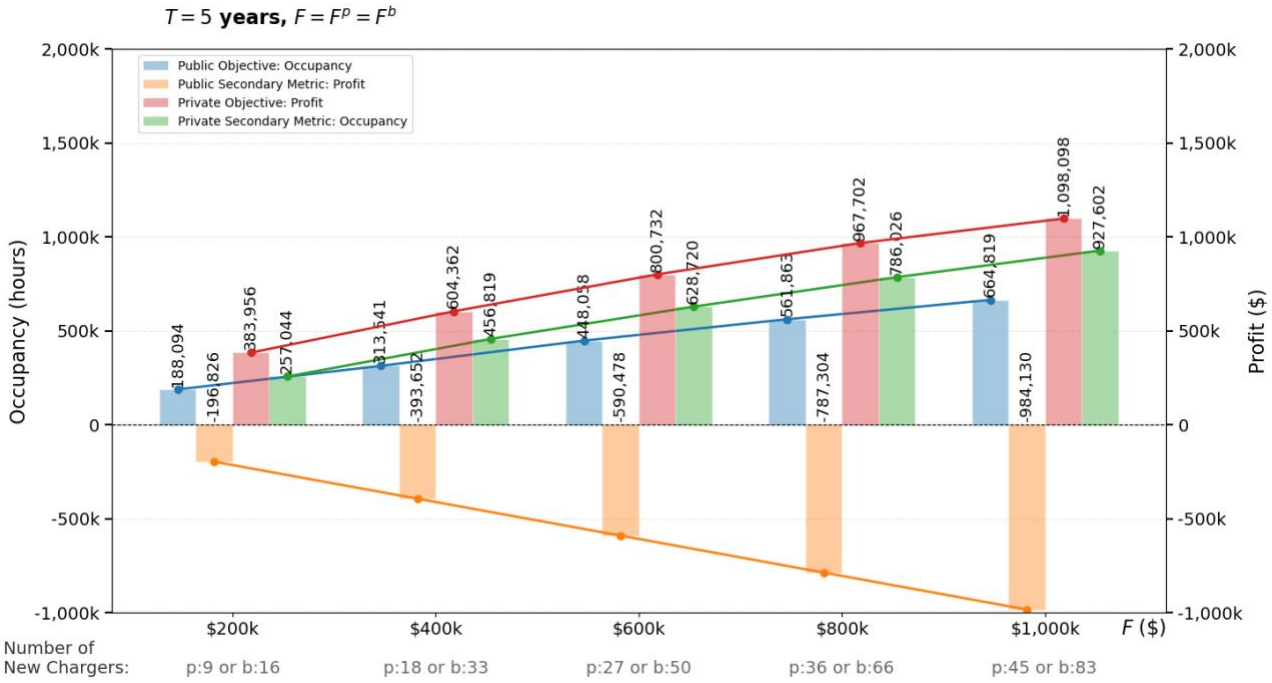


Figure 12. Variations in objectives with different funding

This analysis highlights the critical importance of site selection in charger deployment. For public chargers, occupancy serves as a reliable proxy for site quality. However, for private chargers, identifying suitable quality indicators requires further investigation, as profitability is influenced by both occupancy and business-related factors such as partner capabilities and customer demand.

4.5.2. Impact of planning horizon

To further explore how the performance of public and private chargers evolves over time, we fixed the total funding at $500,000 and varied the planning horizon from 2 to 10 years, in 2-year increments. Figure 13 presents the resulting changes in occupancy and profit for both public- and private-led scenarios. Lines are also used to illustrate trends across planning horizon for easier comparison.

- Private chargers: The profit (red bars) for private chargers grows substantially with time, reaching nearly $1.8 million by Year 10. This strong growth highlights the scalability and long-term revenue potential of private charger deployment, particularly when supported by merchant partners and market demand. Occupancy (green bars) also shows a steady increase, indicating that usage scales well with time. The rate of profit growth exceeds that of occupancy, suggesting an improving revenue-to-cost ratio—likely due to fixed costs being amortized over a longer period.

- Public chargers: Occupancy for public chargers (blue bars) increases steadily with time, but at a slower rate compared to private chargers. Profit (orange bars), however, remains close to $500,000 throughout the entire planning horizon, due to the public budget constraint (Eq. (15)). Specifically, as the planning horizon extends, the operating costs—mainly additional operating costs—continue to rise. To meet this constraint, the funding available for constructing new public chargers decreases, resulting in a reduction in the number of new public stations from 27 in the 2-year planning period

to 17 in the 10-year period. This indicates that, although public chargers become more utilized over the long term, there are clear shortcomings in their construction and operation model under current assumptions, as the number of new chargers declines with an increasing planning horizon.

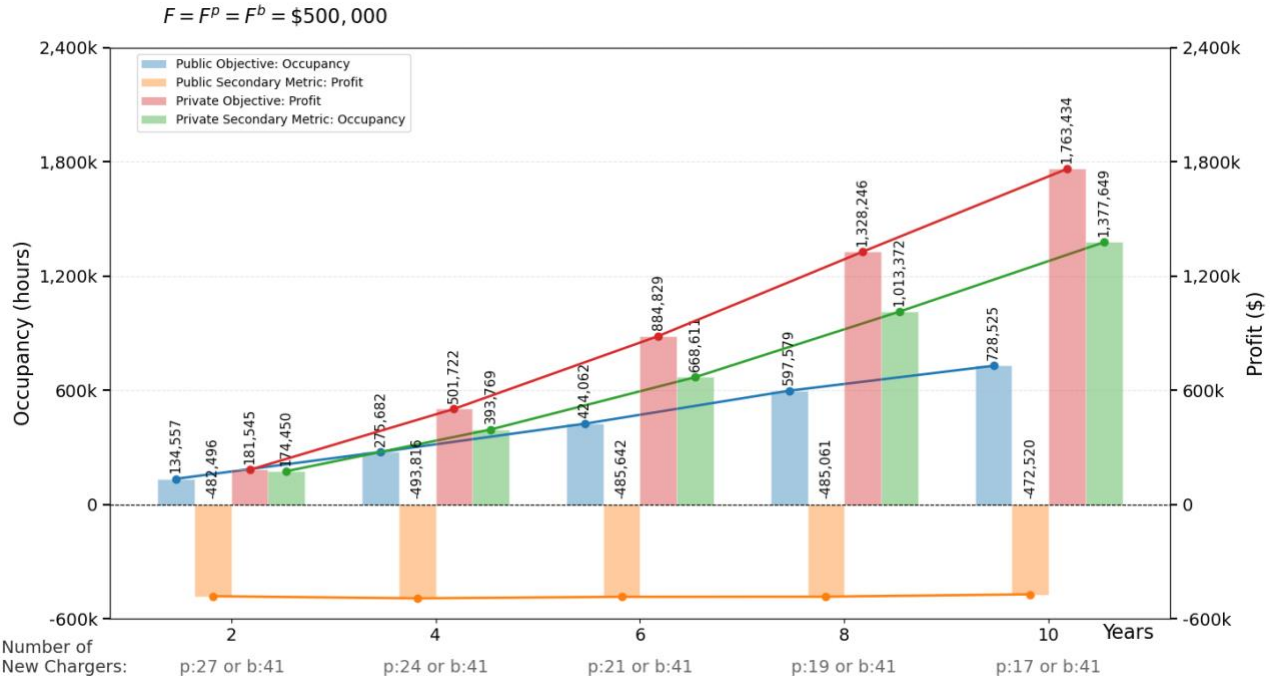


Figure 13. Variations in objectives over time

Figure 14 presents how the components of profit vary across planning horizon. For public chargers, total revenue—i.e., electricity fee income (blue bars)—is charged to cover electricity costs (the portion of the yellow bars equal to the blue bars). The additional operating cost (remaining part of the yellow bars) consistently increases with a longer planning horizon, while the construction cost (orange bars) corresponding decreases. If public chargers are designed to be profitable, the unit electricity fee could be increased, or additional fee items could be introduced; however, it is essential to carefully balance these measures against additional charging cost considerations.

For private chargers, under the current assumptions, subsidies (dark red bars) are provided as direct grants to private investors. As a result, private chargers reach profitability within just two years, since total revenues (a combination of all green bars) consistently exceed operating costs (light red bars) and private budgets (medium red bars). If the subsidies were instead provided as a loan to private investors, the investment payback period would extend from less than two years to four years, which remains a very favorable scenario. This outcome suggests that financial instruments such as loans—rather than direct grants—may be more effective policy tools to support the deployment of private chargers, conserving taxpayer dollars significantly. Furthermore, even when private investors waive overstay parking fees and service fees (the top two green bars) to attract customers, business revenue and energy revenue (the bottom two green bars) are sufficient to maintain profitability around Year 8. This indicates that private investors can flexibly formulate charging strategies based on partner capabilities and customer demand. It is worth

noting that these conclusions are derived under the assumptions of stable growth in EV charging demand and representative ancillary business revenues. Moreover, the analysis does not incorporate several sources of uncertainty in real-world investment environments, including but not limited to demand fluctuations, technology obsolescence due to rapid innovation, regulatory changes, and intensified market competition. Such uncertainties may prolong payback periods and thereby reduce the attractiveness of private investment in charging infrastructure.

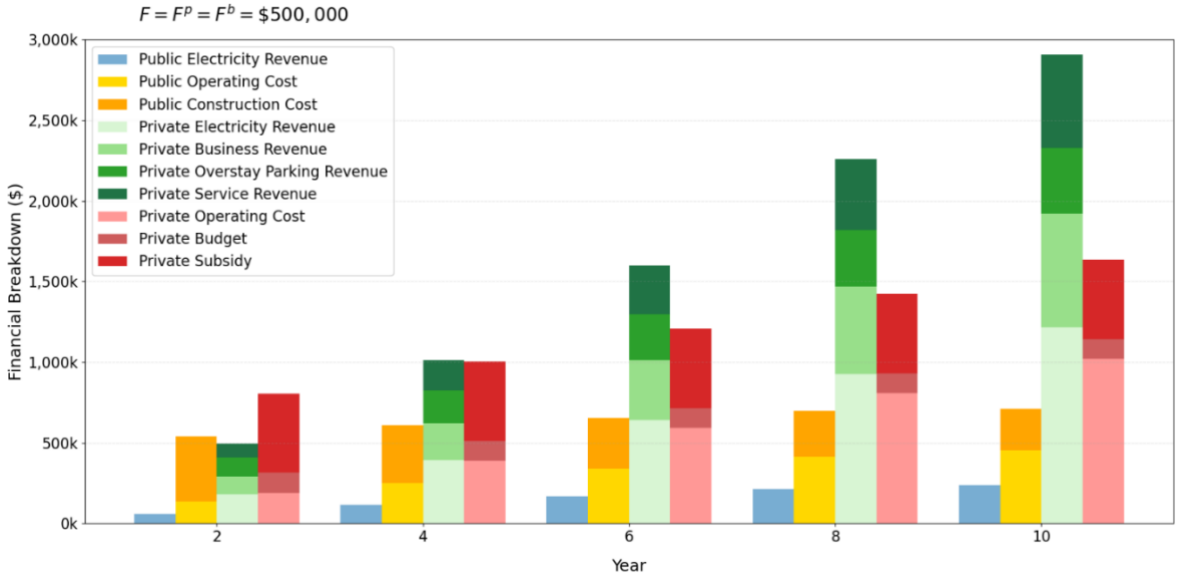


Figure 14. Variations in components of profit over time

This analysis demonstrates that, under fixed funding constraints, deploying private chargers compared to public ones can not only provide more charging services through quantitative advantage, but also generate substantial revenue via private business operations. These benefits, of course, presuppose rational planning and deployment.

4.5.3. Impact of compound annual growth rate for private-led investment

The CAGR ζ of EV adoption scales the multi-year occupancy and profit outcomes. To examine whether the profitability of private-led deployment is robust to uncertainty in adoption forecasts, we vary ζ across $\{0.05, 0.10, 0.15, 0.20\}$, while fixing the budget at \$500,000 over a 5-year horizon, fully allocated to private subsidies. The results are shown in Figure 15.

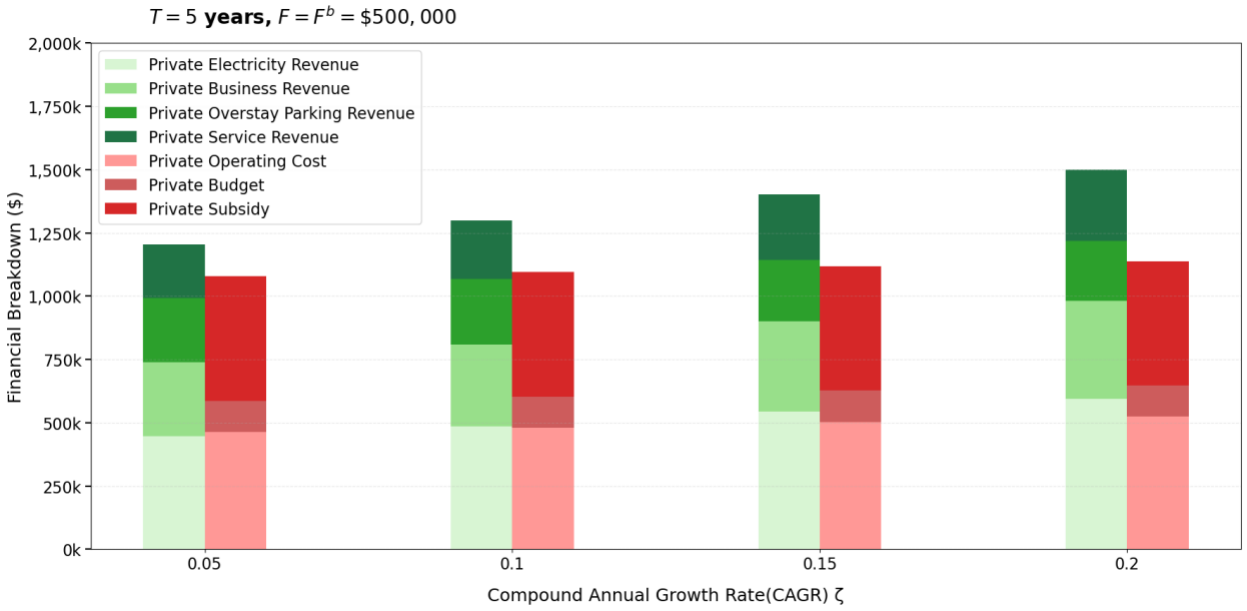


Figure 15. Variations in components of private-led profit over different CAGR

Across all four ζ values, total private revenue exceeds total cost. At the conservative lower bound $\zeta = 0.05$, private-led deployment generates approximately \$619K in profit over five years, with approximately \$127K remaining after the subsidy amount is considered. Within the tested range, no profitability threshold is observed at which private-led deployment becomes unprofitable within the five-year planning horizon. This indicates that the profitability conclusion remains robust even in the face of fluctuations in the EV market under the assumptions of this study.

The revenue decomposition further shows how different revenue components respond to adoption growth. As ζ increases, service fee, electricity, and in-store business revenues increase, while overstay parking revenue declines from \$253K at $\zeta = 0.05$ to \$235K at $\zeta = 0.2$. This pattern can be interpreted through the distinction between session-related and time-based revenue mechanisms. Service-fee, electricity, and in-store business revenues are mainly associated with charging sessions and customer visits, whereas overstay parking revenue depends on post-charging parking duration. Under higher adoption-growth rates, the selected sites appear to support more frequent charging turnover. This increases session-related revenues, but may reduce average post-charging parking time and therefore lower total overstay parking revenue.

4.5.4. Impact of in-store purchase probability for private-led investment

The in-store purchase probability α captures the rate at which charging sessions convert into business transactions at the host establishment, and is the parameter that most directly differentiates private-led from public-led revenue structures. Because consistent data on charging-to-purchase conversion are unavailable, its baseline and tested range are scenario assumptions. We therefore examine its effect by varying α across $\{0, 0.05, 0.10, 0.15\}$, while fixing the budget at \$500,000 over a 5-year horizon, fully allocated to private subsidies.



Figure 16. Variations in components of private-led profit over different in-store purchase probability

Figure 16 reveals that profitability is highly sensitive to α. At $\alpha = 0$, where users charge but make no in-store purchase, total revenue covers operating costs and the government subsidy, but the private investor cannot recover the \$123K capital contribution. Profitability rises rapidly with α: approximately \$172K remains after the subsidy amount is considered at $\alpha = 0.05$, \$348K at the baseline $\alpha = 0.10$, and \$522K at $\alpha = 0.15$. Even at the low conversion probability of $\alpha = 0.05$, private-led deployment still generates profit exceeding the subsidy amount, indicating that the strategy remains financially viable under modest in-store purchase conversion. At the same time, the increase in profit as α rises confirms that ancillary business revenue is an important driver of private-led deployment profitability.

The revenue decomposition further shows how increasing α changes the composition of private revenue. As α increases, overall profit rises steadily, while electricity revenue declines from \$549K to \$485K, service-fee revenue declines from \$261K to \$231K, and overstay parking revenue remains approximately \$242K. This shift occurs because the expected business revenue per charging session, $\alpha \times \pi \times \beta$, increases with α, making the business-revenue component more important in site selection. As a result, locations associated with higher profit per transaction π become more attractive, even if they generate fewer charging sessions and lower charging-related revenue.

These results indicate that the financial attractiveness of private-led deployment depends not only on charging demand but also on whether charging activity can generate complementary business revenue. Therefore, the in-store purchase probability should be carefully calibrated when applying the model to different local business contexts.

5. Conclusions and future work

This study develops a unified CSLP framework to evaluate public-led, private-led, and public-private

coexistence strategies for EV charging infrastructure investment. The upper level represents public agencies and private investors as heterogeneous decision-makers with different objectives, constraints, and performance outcomes, while the lower level predicts station-level utilization under alternative siting decisions. The two levels are iteratively linked through the GISE algorithm, allowing charger deployment decisions and utilization estimates to be updated within a demand-supply coupled planning process.

The Kansas City case study shows that private-led deployment can substantially improve aggregate deployment efficiency under constrained public budgets, but this efficiency gain involves important user-cost and spatial-demographic trade-offs. Under the same $500,000 public budget, private-led deployment leverages private matching capital to support 41 stations, compared with 22 stations under public-led deployment, and expands postal-code coverage from 19 to 28 areas. This finding suggests that, during the current EV market expansion phase, public funds may generate greater aggregate utilization when used to leverage private investment rather than to finance direct public provision alone. However, private-led deployment also increases the cost burden for users dependent on public-access charging. Under Kansas City's conditions, the additional charging cost is approximately $1.40 per occupancy hour, or $193–$484 per year for frequent users, offsetting 24%–37% of typical annual EV fuel-cost savings. In addition, although private-led deployment expands aggregate coverage, it bypasses three postal codes served under public-led planning, including two lower-income, predominantly African American communities. As a case study, Kansas City is used here to demonstrate the framework and illustrate these trade-offs, not to serve as a nationally representative benchmark. The direction of these trade-offs is expected to hold in cities with comparable conditions, while the specific magnitudes reported here would differ with local conditions.

The sensitivity analyses further qualify this policy insight. Results under varying funding levels show that private-led deployment generates higher occupancy and profit as funding increases, but with diminishing returns as high-quality sites become increasingly scarce. Results under varying planning horizons show that private chargers become increasingly profitable over time, while public-led deployment faces growing operating-cost pressure under fixed public funding. Results under varying EV adoption growth rates indicate that private-led profitability remains positive even under conservative market-growth assumptions. However, the in-store purchase probability is critical: when charging users do not make in-store purchases, private investors cannot recover their capital contribution, whereas modest charging-to-purchase conversion makes private-led deployment financially viable. Taken together, these findings indicate that private-led deployment should not be interpreted as a universal replacement for public-led investment. Rather, it can serve as an efficient expansion strategy when paired with safeguards such as fee caps, targeted discounts, differentiated subsidy rates, and set-aside funds for underserved communities.

Several limitations suggest directions for future research. First, the lower-level prediction model is trained only on public charging data and does not include price as a feature, so the private-led advantages that depend on predicted utilization should be read as upper estimates. Future work would benefit from mixed-ownership and price-varying datasets that allow ownership-specific utilization patterns, private-charger behavior, and price sensitivity to be modeled more directly. Second, although the lower-level model captures station-level utilization responses through contextual features and historical charging records, it does not explicitly model individual-level charging choice mechanisms. Future extensions could integrate

discrete-choice, agent-based, or network equilibrium models with the ML-based predictor to support more detailed evaluation of user responses under alternative behavioral and access conditions. Finally, the framework could be expanded to incorporate power-grid constraints, charging-speed heterogeneity, and regional coordination for larger-scale infrastructure planning.

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